Rough Paths

Hölder spaces

Basic properties

Increment of a path : $X : [0,T] \longrightarrow \mathbb{R}^d$, $s,t \in [0,T]$, $X_{s,t} := X_t - X_s$

Space of continuous path : $C := C([0,T];\mathbb{R}^d) := \{ X : [0,T] \longrightarrow \mathbb{R}^d \text{ continuous} \}$ and we

write $\|X\|_\infty = \sup_{t\in[0,T]} |X_t|$ for the supremum norm

α-Hölder seminorm : $X : [0,T] \longrightarrow \mathbb{R}^d$, $\|X\|_\alpha := \sup_{0\le s<t\le T} \frac{|X_{s,t}|}{|t-s|^\alpha}$, $\alpha \in (0,1]$

If $\|X\|_\alpha < +\infty \implies X$ is uniformly continuous $\implies$ it makes sense to call

the space of α-hölder continuous path $:= C^\alpha([0,T];\mathbb{R}^d) := \{ X : [0,T] \to \mathbb{R}^d : \|X\|_\alpha < +\infty \}$

$\|X\|_\alpha$ is a seminorm :

- $\|cX\|_\alpha = |c|\,\|X\|_\alpha$
- $\|X+Y\|_\alpha \le \|Y\|_\alpha + \|X\|_\alpha$
- $\|X\|_\alpha = 0 \not\Rightarrow X = 0$ infact for every constant path $\|X\|_\alpha = 0$

Proof

- $\sup_{0\le s<t\le T} \frac{|cX_{s,t}|}{|t-s|^\alpha} = c \sup_{0\le s\le t\le T} \frac{|X_{s,t}|}{|t-s|^\alpha}$
- $\sup_{0\le s\le t\le T} \frac{|X_{s,t} + Y_{s,t}|}{|t-s|^\alpha} \le \sup_{0\le s\le t\le T} \left(\frac{|X_{s,t}|}{|t-s|^\alpha} + \frac{|Y_{s,t}|}{|t-s|^\alpha} \right) \le \sup_{0\le s\le t\le T} \frac{|X_{s,t}|}{|t-s|^\alpha} + \sup_{0\le s\le t\le T} \frac{|Y_{s,t}|}{|t-s|^\alpha}$
- if $X = x_0 \implies X_{s,t} = 0 \quad \forall\, s<t$ in $[0,T] \implies \|X\|_\alpha = 0$ $\square$

$\|\cdot\|_\alpha$ is lower semi-continuous : let $\alpha \in (0,1]$ and $X : [0,T] \longrightarrow \mathbb{R}^d$. Let $(X^n)_{n\ge1} \subseteq C^\alpha$ be a sequence of α-Hölder continuous paths and assume $X^n \longrightarrow X$ pointwise. Then

$$\|X\|_\alpha \le \liminf_{n\to+\infty} \|X^n\|_\alpha$$

Proof : $\forall\, 0\le s<t\le T$ we have:

$$\frac{|X_{s,t}|}{|t-s|^\alpha} = \liminf_n \frac{|X^n_{s,t}|}{|t-s|^\alpha} \le \liminf_n \sup_{\substack{s,t\in[0,T]\\ s<t}} \frac{|X^n_{s,t}|}{|t-s|^\alpha} = \liminf_n \|X^n\|_\alpha$$

, taking the sup we have the th. $\square$

$C^\alpha([0,T];\mathbb{R}^d)$ as Banach space: we can equip $C^\alpha([0,T];\mathbb{R}^d)$ with the norm:

$$X \longmapsto |X_0| + \|X\|_\alpha$$

Equipped with this norm $C^\alpha([0,T];\mathbb{R}^d)$ is a Banach space not separable.

Proof: • Norm: the first two properties as above; if $|X_0| + \|X\|_\alpha = 0 \Rightarrow |X_0| = 0 \wedge \|X\|_\alpha = 0$

so X_t starts at zero and it has no increments $\Rightarrow X \equiv 0$.

• Banach space: let $(X^n)_n$ Cauchy sequence.

Step 1: find a X s.t. $X^n \longrightarrow X$ pointwise:

$\forall t \in [0,T]$, we want to see that $(X_n^t)_n$ is Cauchy in $\mathbb{R}^d$ so $\exists \overline{X}_t := \lim_n X_n^t$

$$|X_t^n - X_t^m| = |X_t^n - X_0^n + (X_0^n - X_0^m) + X_0^m - X_t^m| =$$

$$\leq |X_0^n - X_0^m| + |(X_t^n - X_t^m) - (X_0^n - X_0^m)|$$

$$\leq |X_0^n - X_0^m| + \|X^n - X^m\|_\alpha t^\alpha \xrightarrow[n,m\to\infty]{} 0$$

Step 2: $\overline{X}_t \in C^\alpha([0,T];\mathbb{R}^d)$:

$$\|\overline{X}\|_\alpha \leq \liminf_n \|X^n\|_\alpha \leq \underbrace{\sup_n \|X^n\|_\alpha < +\infty}_{\text{Cauchy} \Rightarrow \text{bounded}}$$

Step 3: $X^n \longrightarrow \overline{X}$ in the C^α-norm:

$$\underbrace{|X_0^n - \overline{X}_0|}_{\substack{\text{this goes to}\\ 0 \text{ as } n\to+\infty\\ \text{by def.}}} + \underbrace{\|X^n - \overline{X}\|_\alpha}$$

$$\frac{|X_{s,t}^n - \overline{X}_{s,t}|}{|t-s|^\alpha} = \lim_m \frac{|X_{s,t}^n - \overline{X}_{s,t}^m|}{|s-t|^\alpha} \leq \limsup_m \|X^n - \overline{X}^m\|_\alpha$$

, taking the $\sup_{s<t}(\cdot)$ and the $\lim_n$ we have the thesis

• Not separable: the idea is to construct $\{f_E : E \subseteq \mathbb{N}\}$ not countable s.t. $d(f_E, f_F) \geq 1$ (given by the norm) $\forall E \neq F$ in $\mathbb{N}$. In this way cannot exist a dense countable set (obviously). □

Equivalent norm: Since $[0,T]$ is compact the map:

$$X \longmapsto \|X\|_\infty + \|X\|_\alpha$$

is an equivalent norm of $X \longmapsto |X_0| + \|X\|_\alpha$

Proof: $|X_0| + \|X\|_\alpha \leq \|X\|_\infty + \|X\|_\alpha$ obvious. But $\|X\|_\infty \leq |X_0| + T^\alpha \|X\|_\alpha \Rightarrow \|X\|_\infty + \|X\|_\alpha \leq (1+T^\alpha)(|X_0| + \|X\|_\alpha)$ □

The case $\alpha > 1$ is not interesting : let $X \in C^\alpha$ for some $\alpha > 1$. Show the path must be equal to a constant.

Proof : We want to see the $|X_{0,t}| = 0$ $\forall t \in [0,T]$ hence $X_0 = X_t$. The idea is to have $|X_{0,t}| \leq \text{something}^{\alpha-1}$ so that $\alpha - 1 > 1$ and when something goes to zero we have conv:

- $\Pi = (s_i)_{i=1,\dots,N}$ a partition of $[0,t]$
- $|X_{0,t}| = \left| \sum_{i=0}^{N-1} X_{s_i, s_{i+1}} \right| \leq \sum_{i=0}^{N-1} |X_{s_i, s_{i+1}}| \leq \|X\|_\alpha \sum_{i=0}^{N-1} |s_{i+1} - s_i|^\alpha$

$$\leq \|X\|_\alpha \left(\max_{0 \leq i < N} |s_{i+1} - s_i|^{\alpha-1} \right) \sum_{i=0}^{N-1} \underbrace{|s_{i+1} - s_i|}_{= s_{i+1} - s_i} = \|X\|_\alpha \overset{\text{the mesh}}{|\Pi|}^{\alpha-1} t \xrightarrow{|\Pi| \to 0} 0$$

$\square$

The inclusion $C^\beta \subset C^\alpha$: if $0 < \alpha < \beta \leq 1$ then $C^\beta \subsetneq C^\alpha$. In particular $C^\beta \hookrightarrow C^\alpha$ is an embedding.

Proof $\subseteq$: $\frac{|X_{s,t}|}{|t-s|^\alpha} = \frac{|X_{s,t}|}{|t-s|^\beta} |t-s|^{\beta-\alpha} \leq \frac{|X_{s,t}|}{|t-s|^\beta} T^{\beta-\alpha} \Rightarrow \|X\|_\alpha \leq \|X\|_\beta T^{\beta-\alpha}$

the inequality gives the embedding property

$\not\supseteq$: $X_t = t^\alpha$: since $|t^\alpha - s^\alpha| \leq |t-s|^\alpha$, $\|X\|_\alpha \leq 1$ but $\frac{|t^\alpha - 0|}{|t-0|^\beta} = t^{\alpha-\beta} \xrightarrow[t \to 0]{} +\infty$

$\square$

Interpolation : let $0 < \alpha < \beta \leq 1$ and let $X : [0,T] \to \mathbb{R}^d$. Then

$$\|X\|_\alpha \leq \|X\|_\beta^{\frac{\alpha}{\beta}} \left(\sup_{0 \leq s < t \leq T} |X_{s,t}| \right)^{1-\frac{\alpha}{\beta}}$$

Proof $\frac{|X_{s,t}|}{|t-s|^\alpha} = \left(\frac{|X_{s,t}|}{|t-s|^\beta} \right)^{\frac{\alpha}{\beta}} |X_{s,t}|^{1-\frac{\alpha}{\beta}}$, taking the $\sup_{s<t}$ and using $\sup AB \leq \sup A \sup B$ (A, B positives). $\square$

Uniform convergence and $\|\cdot\|_\alpha$: let $0 < \alpha < \beta \leq 1$. $X \in C$ continuous path. Let $(X^n)_{n \geq 1} \subset C^\beta$ and $\sup_n \|X^n\|_\beta < +\infty$. If $X^n \to X$ uniformly then $X \in C^\beta$ and $\|X^n - X\|_\alpha \xrightarrow{n \to +\infty} 0$ (and so in the α-Hölder norm).

Proof : · $X \in C^\beta$: $\|X\|_\beta \underset{\text{lower-semi c.}}{\leq} \liminf_n \|X\|_\beta \leq \sup_n \|X^n\|_\beta < \infty$

· $\|X^n - X\|_\alpha \underset{\text{interpolation}}{\leq} \|X^n - X\|_\beta^{\frac{\alpha}{\beta}} \Big(\underbrace{\sup_{0 \leq s \leq t \leq T} |X^n_{s,t} - X_{s,t}|}_{\to 0 \text{ as } n \to +\infty \text{ because } \|X^n - X\|_\infty \to 0} \Big)^{1-\frac{\alpha}{\beta}}$

Now $\| X^n - X \|_\beta \leq \sup_n \|X^n\|_\beta + \|X\|_\beta < +\infty$ so as $n \longrightarrow +\infty$

previous point

$\| X^n - X \|_\alpha \longrightarrow C \cdot 0 = 0.$

$\square$

Remark: · Uniform convergence $\not\Rightarrow$ Hölder-convergence $\left(f_n(x) = \frac{1}{n} \sin(n^{\frac{1}{\alpha}} x) \right.$ on $[0, 2\pi]$; $\|f_n\|_\infty = \frac{1}{n}$

so $f_n \xrightarrow{\text{uniformly}} 0$ but $\left. \|f_n\|_\alpha \not\to 0 \right)$

· Hölder convergence $\Rightarrow$ Uniform convergence (by def. and the equivalent norm)

· Hölder-seminorm convergence $\not\Rightarrow$ uniform convergence but implies $\| X_n - X \|_\infty \longrightarrow$ constant

Compactness: $0 < \alpha < \beta \leq 1$. $(X^n)_{n \geq 1} \subset C^\beta$ with $\underbrace{\sup_{n \geq 1} \left(|X_0^n| + \|X^n\|_\beta \right)}_{\text{bounded in the } C^\beta\text{-norm}} < \infty$. Then

$\exists X \in C^\beta$ and $\exists (n_k)_{k \geq 1}$ s.t. $X_{n_k} \longrightarrow X$ in the C^α-norm.

Proof

$(X^n)_n$ is uniformly bounded $\left(\|X^n\|_\infty \leq \|X^n\|_\infty + \|X^n\|_\beta \lesssim |X_0^n| + \|X^n\|_\beta \leq C \quad \forall n \right)$

$(X^n)_n$ is uniformly continuous $\left(|X_t^n - X_s^n| \leq \|X^n\|_\beta |t-s|^\beta \leq C |t-s|^\beta \Rightarrow \text{equicontinuous} \right)$

So using Ascoli-Arzelà $\exists n_k$ s.t. $X_{n_k} \xrightarrow{\text{uniformly}} X$. For the previous result we have the th.

$\square$

Corollary: $0 < \alpha < \beta \leq 1$, $i : C^\beta \hookrightarrow C^\alpha$ is a compact embedding i.e. $\forall K$ bounded in

C^β, K is relatively compact in C^α (i.e. $\overline{K}$ compact in C^α and since is metrizable

equivalent to ask $\forall$ sequence in $\exists$ subsequence that converges in the C^α-norm). $\leftarrow$ Straight forward from the previous result

The closure of smooth paths

Closure of smooth paths: for $\alpha \in (0,1]$, $C^{0,\alpha} := C^{0,\alpha}([0,T]; \mathbb{R}^d)$ it the closure

of the space of smooth paths from $[0,T] \to \mathbb{R}^d$ w.r.t. the α-Hölder norm.

$C^{0,\alpha}$ is a Banach space: $C^{0,\alpha}([0,T]; \mathbb{R}^d)$ equipped with the α-Hölder norm is a Banach

space.

$$\|X\|_\alpha = \sup_{0 \leq s \leq t \leq T} \frac{|X_{s,t}|}{|t-s|^\alpha} \leq \sup_{t \in [0,T]} |X'(t)| \, T^\alpha = \mathrm{Lip}(X) \, T^\alpha$$

Proof: $\{ \text{smooth paths from } [0,T] \to \mathbb{R}^d \} \subseteq C^\alpha([0,T]; \mathbb{R}^d)$ and they are a vector subspace.

But the closure of a vector subspace is a vector subspace and a closed vector subspace of a Banach is obviously a Banach.

$\square$

Characterization of $C^{0,\alpha}$ for $\alpha \in (0,1)$: let $\alpha \in (0,1)$ and $X : [0,T] \to \mathbb{R}^d$ be a path. Then

$$X \in C^{0,\alpha} \iff \lim_{\delta \to 0} \sup_{|t-s|<\delta} \frac{|X_{s,t}|}{|t-s|^\alpha} = 0$$

Proof: $\boxed{\Rightarrow}$ let $\varepsilon > 0$. By def. of closure $\exists Y \in C^\infty$ s.t. $\|X - Y\|_\alpha < \frac{\varepsilon}{2}$.

Since Y is smooth, it's Lip. on $[0,T]$ for some constant L. We choose δ small enough s.t. $L\delta^{1-\alpha} < \frac{\varepsilon}{2}$ (here we use $\alpha < 1$), so for $|t-s| < \delta$ we have:

$$\frac{|X_{s,t}|}{|t-s|^\alpha} \le \frac{|X_{s,t} - Y_{s,t}|}{|t-s|^\alpha} + \underbrace{\frac{|Y_{s,t}|}{|t-s|^\alpha}}_{\le \frac{L|t-s|}{|t-s|^\alpha} \le L\delta^{1-\alpha}} \le \|X - Y\|_\alpha + L\delta^{1-\alpha} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

So $\forall \varepsilon > 0$, choosing δ small enough we have that $\sup_{|t-s|<\delta} \frac{|X_{s,t}|}{|t-s|^\alpha} < \varepsilon$. hence the th. holds.

$\boxed{\Leftarrow}$ Let $\varepsilon > 0$, by assumption it exists a $\delta > 0$ s.t. $\sup_{|t-s|<\delta} \frac{|X_{s,t}|}{|t-s|^\alpha} < \varepsilon$. $(*_1)$ We can assume that this $\delta \le 1$.

Step 1: Build a piecewise-linear approximation of X:

Let $\pi = \{0 = u_0 < u_1 < \dots < u_N = T\}$ partition of $[0,T]$ with equidistant endpoints (i.e. $u_i = i\frac{T}{n}$), s.t. $|\pi| = |u_{i+1} - u_i| < \delta$.

The linear approximation of X is then a $Y^\ell : [0,T] \to \mathbb{R}^d$ s.t. $Y(u_i) = X(u_i)$ and linear on $[u_i, u_{i+1}]$ $\forall i$. In particular a linear approximation is Lip. cont. on $[0,T]$ with constant:

$$\max_{0 \le i < N} \frac{|X_{u_i, u_{i+1}}|}{|u_{i+1} - u_i|} = \frac{1}{|\pi|} \max_{0 \le i < N} |X_{u_i, u_{i+1}}|.$$

Step 2: Buid Y a C^∞-approximation of X:

Given Y^ℓ we can smooth out in a small neighborhood of each of the points of u_i s.t. we alway have $Y(u_i) = X(u_i)$ and the lip. constant increases by an arbitrary amount; in this case we want $Y : [0, T] \longrightarrow \mathbb{R}^d$ smooth obtained by smoothing out Y^ℓ s.t.:

- $Y_{u_i} = X_{u_i} \quad \forall i$
- the Lip constant of L is s.t. $L \leq \left(\frac{1}{|\pi|} \max_{0 \leq i < N} |X_{u_i, u_{i+1}}| \right) + \mathcal{E}$, in particular

$$\frac{1}{|\pi|} \max_{0 \leq i < N} |X_{u_i, u_{i+1}}| = \frac{1}{|\pi|^{1-\alpha}} \max_{0 \leq i < N} \frac{|X_{u_i, u_{i+1}}|}{|u_{i+1} - u_i|} \underset{|u_{i+1}-u_i| = |\pi| < \delta \Rightarrow (*_1)}{\leq} \frac{1}{|\pi|^{1-\alpha}} \mathcal{E} \implies L \leq \mathcal{E} \left(\frac{1}{|\pi|^{1-\alpha}} + 1 \right) \quad (*_2)$$

<u>Step 3</u>: See that Y is close enough to X in the C^α-norm:

Now the C^α-norm of $X - Y$ is:

$$(*_3) \quad \underbrace{|X_0 - Y_0|}_{= 0 \text{ by constr.}} + \sup_{s<t} \frac{|X_{s,t} - Y_{s,t}|}{|t-s|^\alpha} = \max \left(\sup_{\substack{s<t \\ j=K}} \frac{|X_{s,t} - Y_{s,t}|}{|t-s|^\alpha} ; \sup_{\substack{s<t \\ j \neq K}} \frac{|X_{s,t} - Y_{s,t}|}{|t-s|^\alpha} \right)$$

given $0 \leq s < t \leq T$ $\exists!$ j, K s.t. $s \in [u_j, u_{j+1})$ and $K \in (u_K, u_{K+1}]$

s,t are in the same interval

s,t are in different interval

But if $j = K$:

$$\cdot \frac{|X_{s,t}|}{|t-s|^\alpha} \underset{\substack{|t-s| < |\pi| < \delta, \\ \text{we use } (*_1)}}{<} \mathcal{E} \qquad \cdot \frac{|Y_{s,t}|}{|t-s|^\alpha} \leq L |t-s|^{1-\alpha} \underset{(*_2)}{\leq} \mathcal{E} \left(\frac{1}{|\pi|^{1-\alpha}} + 1 \right) |t-s|^{1-\alpha} \underset{|t-s| \leq |\pi|}{\leq}$$

$$\leq \mathcal{E} \left(1 + |t-s|^{1-\alpha} \right) \underset{|t-s|^{1-\alpha} \leq |\pi|^{1-\alpha} \leq 1}{\leq} 2\mathcal{E}$$

$$\text{So} \quad \sup_{\substack{s<t \\ j=K}} \frac{|X_{s,t} - Y_{s,t}|}{|t-s|^\alpha} \leq \sup_{\substack{s<t \\ j=K}} \frac{|X_{s,t}|}{|t-s|^\alpha} + \frac{|Y_{s,t}|}{|t-s|^\alpha} = 3\mathcal{E}$$

If $j \neq K$: using that $X_{u_{j+1}} = Y_{u_{j+1}}$ and $X_{u_K} = Y_{u_K}$ we have

$$X_{s,t} - Y_{s,t} = X_{s, u_{j+1}} - Y_{s, u_{j+1}} + X_{u_K, t} - Y_{u_K, t}$$

$$\implies \frac{|X_{s,t} - Y_{s,t}|}{|t-s|^\alpha} \leq \frac{|X_{s,u_{j+1}}|}{|t-s|^\alpha} + \frac{|Y_{s,u_{j+1}}|}{|t-s|^\alpha} + \frac{|X_{u_K,t}|}{|t-s|^\alpha} + \frac{|Y_{u_K,t}|}{|t-s|^\alpha} \leq 6\mathcal{E}$$

on them $|u_{j+1} - s| < |\pi| < \delta$, $|u_K - t| < |\pi| < \delta$ and so I can use $(*_1)$ as before to obtain $\leq \mathcal{E}$

on them it is true that $|u_{j+1} - s| < |t-s|$ and $|t - u_K| < |t-s|$ that is the only thing used in the prev. estimate to have $\leq 2\mathcal{E}$

So $(*_3) \leq 6\mathcal{E}$, since $\mathcal{E}$ is arbitrary we have $X \in C^{0,\alpha}$.

$\square$

$C^{0,\alpha} \subsetneq C^{\alpha}$ for $\alpha \in (0,1)$: let $\alpha \in (0,1)$ and $X : [0,T] \to \mathbb{R}$ be the path $X_t = t^{\alpha}$. Then $X \in C^{\alpha}$ but $X \notin C^{0,\alpha}$.

Proof $\cdot X \in C^{\alpha}$: $\frac{|t^{\alpha} - s^{\alpha}|}{|t-s|^{\alpha}} \leq 1 \Rightarrow |X|_{C^{\alpha}} < 1$. But

$$\lim_{\delta \to 0} \sup_{|t-s|<\delta} \frac{|t^{\alpha}-s^{\alpha}|}{|t-s|^{\alpha}} \underset{\substack{s=0 \\ t<\delta}}{\geq} \lim_{\delta \to 0} \frac{t^{\alpha}}{t^{\alpha}} = 1$$ and so by the charac. of $C^{0,\alpha}$, $X \notin C^{0,\alpha}$.

$C^{\beta} \subset C^{0,\alpha}$: let $0 < \alpha < \beta \leq 1$. Then $C^{\beta} \subset C^{0,\alpha}$.

Proof : let $|t-s| < \delta$. Then $\frac{|X_{s,t}|}{|t-s|^{\alpha}} = \frac{|X_{s,t}|}{|t-s|^{\beta}} |t-s|^{\beta-\alpha} \leq \|X\|_{\beta} \, \delta^{\beta-\alpha} \Rightarrow$ for the charac. of $C^{0,\alpha}$ we have the thesis. $\square$

Recap : for $0 < \alpha < \beta \leq 1$ we have: $C^{\beta} \subsetneq C^{0,\alpha} \subsetneq C^{\alpha}$

for example $X_0 := 0$ $X_t := \frac{t^{\alpha}}{1+|\log t|}$ $(t>0)$

The case $\alpha = 1$: $C^{0,1}([0,T], \mathbb{R}^d) = C^1([0,T], \mathbb{R}^d)$, i.e. the continuosly differentiable path. In particular, also in this case, $C^{0,1}([0,T], \mathbb{R}^d) \subsetneq C^1([0,T], \mathbb{R}^d)$

just take $X_t = |t - \underbrace{\tfrac{T}{2}}|$, is Lip but not differentiable

to center the discontinuity in $\frac{T}{2}$

Proof The strategy is to see that the smooth paths are dense in $C^1([0,T], \mathbb{R}^d)$ w.r.t. the C^1-norm $(|X_0| + \|X\|_1)$ so that the closure $C^{0,1} = C^1$.

First of all, we want to use a more useful norm :

Step 1 : for the smooth paths (it needs only C^1), $\|X\|_1 = \sup_{t \in [0,T]} |\dot{X}_t|$:

$\boxed{\leq}$: $X_t - X_s = \int_s^t \dot{X}_u \, du \Rightarrow |X_t - X_s| \leq |t-s| \sup_{t \in [0,T]} |\dot{X}_t|$

$\boxed{\geq}$: $\frac{|X_{t+h} - X_h|}{|h|} \leq \|X\|_1$, taking the $\lim_{h \to 0}$ and the $\sup_{t \in [0,T]}$ we have what we want.

Step 2 : C^{∞} is dense in C^1 w.r.t. the norm $|X_0| + \sup_{t \in [0,T]} |\dot{X}_t|$:

Fix $X \in C^1$, so $\dot{X} \in C([0,T] ; \mathbb{R}^d)$ and so (since $[0,T]$ is compact) $\exists \, g_n \in C^{\infty}$

for example with convolution of mollifiers

s.t. $g_n \longrightarrow \dot{X}$ uniformly in $[0,T]$. Set $Y_n^t := X_0 + \int_0^t g_n(u) \, du$ and the th. follows straightforward $\square$

Two parameters function

$\leftarrow$ it can be E finite dim. vec. space and all the results of this section hold

$C_2(\Delta_{[0,T]};\mathbb{R}^{d\times d})$: let $\Delta_{[0,T]} := \{(s,t)\in[0,T] : s\le t\}$, we denote

$$C_2 := C_2(\Delta_{[0,T]};\mathbb{R}^d\times\mathbb{R}^d) := \{A : \Delta_{[0,T]} \longrightarrow \mathbb{R}^{d\times d} : A \text{ continuous}\}$$

$C_2^\alpha(\Delta_{[0,T]};\mathbb{R}^{d\times d})$: $C_2^\alpha(\Delta_{[0,T]};\mathbb{R}^{d\times d}) = \{A\in C_2(\Delta_{[0,T]};\mathbb{R}^{d\times d}) : \|A\|_\alpha<\infty\}$ where

$$\|A\|_\alpha := \sup_{0\le s<t\le T}\frac{|A_{s,t}|}{|t-s|^\alpha}$$

$\leftarrow$ new $A_{s,t} := A(s,t)$

$\leftarrow$ penso prendiamo $|M| = \sum_{i,j} M_{ij}^2$ così $|s\otimes t| = |s|\,|t|$ in $\mathbb{R}^d\times\mathbb{R}^d$

Remark: • if $A\in C_2^\alpha \implies |A_{s,t}|\le\|A\|_\alpha|t-s|^\alpha \implies \underset{\substack{\uparrow\\ A\text{ is cont.}}}{|A_{t,t}| =} \lim_{s\to t}|A_{s,t}|\le 0$

$\implies$ A has to be 0 on the diagonal.

• if $\alpha>1$, A has not be constant (as for the path) but its contribution on finite partition $\xrightarrow{|\pi|\mapsto 0} 0$:

$$\sum_{[s,t]\in\pi}|A_{s,t}|\le\|A\|_\alpha\sum_{[s,t]\in\pi}|t-s|^\alpha\le\|A\|_\alpha\Big(\sum_{[s,t]\in\pi}|t-s|\Big)|\pi|^{\alpha-1} = \|A\|_\alpha\,T\,|\pi|^{\alpha-1}.$$

$C_2^\alpha(\Delta_{[0,T]};\mathbb{R}^{d\times d})$ is Banach: $\alpha\in(0,1]$; C_2^α with $\|\cdot\|_\alpha$ is a Banach space.

Proof $\|\cdot\|_\alpha$ is a norm: the Δ inequ. and the product for scalar are obvious; if

$\|A\|_\alpha = 0 \implies |A_{s,t}| = 0 \quad \forall s\ne t$ but we already know that $A_{t,t}=0$.

Complete: basically the same argument used for C^α.

□

$C^\alpha([0,T];\mathbb{R}^d)\times C_2^\beta(\Delta_{[0,T]};\mathbb{R}^d\times\mathbb{R}^d)$ is Banach: given $\alpha,\beta\in(0,1]$, we define the norm on

$C^\alpha\times C_2^\beta$ given by:

$$(X,A)\longmapsto \underbrace{X_0+\|X\|_\alpha}_{\text{norm in } C^\alpha} + \underbrace{\|A\|_\beta}_{\text{norm in } C_2^\beta}$$

Proof: the product of two Banachs is always Banach with norm the sum of the norms (or the max, they are equivalent).

□

As for the paths, we have the same results of lower semicontinuity and interpolation for the 2-param. function as well:

$\|\cdot\|_\alpha$ in C_2^α is lower semi-continuous : let $(A^n)_n \subseteq C_2^\alpha(\Delta_{[0,T]}; \mathbb{R}^d \times \mathbb{R}^d)$ and assume $A^n \longrightarrow A$ pointwise. Then

$$\|A\|_\alpha \leq \liminf_{n \to +\infty} \|A^n\|_\alpha$$

Proof : $\forall\, 0 \leq s < t \leq T$ we have:

$$\frac{|A_{s,t}|}{|t-s|^\alpha} = \liminf_n \frac{|A^n_{s,t}|}{|t-s|^\alpha} \leq \liminf_n \sup_{\substack{s,t \in [0,T] \\ s<t}} \frac{|A^n_{s,t}|}{|t-s|^\alpha} = \liminf_n \|A^n\|_\alpha,$$ taking the sup we have the th. $\square$

Interpolation in C_2^α : let $0 < \alpha < \beta$ and let $A : \Delta_{[0,T]} \longrightarrow \mathbb{R}^d \times \mathbb{R}^d$. Then

$$\|A\|_\alpha \leq \|A\|_\beta^{\frac{\alpha}{\beta}} \left(\sup_{0 \leq s < t \leq T} |A_{s,t}| \right)^{1-\frac{\alpha}{\beta}}$$

Proof $$\frac{|A_{s,t}|}{|t-s|^\alpha} = \left(\frac{|A_{s,t}|}{|t-s|^\beta} \right)^{\frac{\alpha}{\beta}} |A_{s,t}|^{1-\frac{\alpha}{\beta}}$$, taking the $\sup_{s<t}$ and using $\sup AB \leq \sup A \sup B$ (A, B positives). $\square$

The space of Rough paths

Basic definitions

<u>α - Hölder rough path</u> : given $\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right]$, an α-Hölder rough path over $\mathbb{R}^d$ is a pair $\mathbf{X} := (X, \mathbb{X})$ in $C^\alpha \times C_2^{2\alpha}$ s.t. the Chen's relation holds $\forall\ 0 \le s \le u \le t \le T$:

↑ lift of X

$$\mathbb{X}_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t} \quad , \text{ in coordinate}$$

$$\mathbb{X}^{i,j}_{s,t} = \mathbb{X}^{i,j}_{s,u} + \mathbb{X}_{u,t} + X^i_{s,u}\, X^j_{u,t} \qquad \forall\ i,j = 1, \dots, d$$

We denothe this space $\mathscr{C}^\alpha([0,T]; \mathbb{R}^d)$ equipped with the $C^\alpha \times C^{2\alpha}$ Banach-norm

$$\mathbf{X} \longmapsto |X_0| + |||\mathbf{X}|||_\alpha \quad , \text{ where } \quad |||\mathbf{X}|||_\alpha = \| X \|_\alpha + \| \mathbb{X} \|_{2\alpha} .$$

The problem is that the Chen's relation is not linear so $\mathscr{C}^\alpha([0,T]; \mathbb{R}^d)$ is not even a vector space. Although this, $\mathscr{C}^\alpha$ is complete w.r.t. the metric induced by the norm:

<u>$\mathscr{C}^\alpha([0,T]; \mathbb{R}^d)$ is complete</u> : $\mathscr{C}^\alpha$ with the metric:

$$(\mathbf{X}, \tilde{\mathbf{X}}) \longmapsto |X_0 - \tilde{X}_0| + \| \mathbf{X} ; \tilde{\mathbf{X}} \|_\alpha \quad \text{where}$$

$$\| \mathbf{X} ; \tilde{\mathbf{X}} \|_\alpha := \| X - \tilde{X} \|_\alpha + \| \mathbb{X} - \tilde{\mathbb{X}} \|_{2\alpha} \quad (\text{it is the associate pseudometric})$$

is complete.

<u>Proof</u> : taking a Cauchy sequence $\mathbf{X}^n = (X^n, \mathbb{X}^n)$ in $\mathscr{C}^\alpha$, since $C^\alpha \times C^{2\alpha}$ is Banach, $\exists\ \mathbf{X} \in \mathscr{C}^\alpha$ s.t. $\mathbf{X}^n \longrightarrow \mathbf{X}$ in the $C^\alpha \times C^{2\alpha}$-norm. We want to see that $\mathbf{X}$ satisfy the Chen's equality. So:

$\forall n$ holds so $\mathbb{X}^n_{s,u} - \mathbb{X}^n_{s,t} - \mathbb{X}^n_{t,u} - X^n_{s,t} \otimes X^n_{t,u} = 0$

$$| \mathbb{X}_{s,u} - \mathbb{X}_{s,t} - \mathbb{X}_{t,u} - X_{s,t} \otimes X_{t,u} | \le$$

$$\le \underbrace{| \mathbb{X}_{s,u} - \mathbb{X}^n_{s,u} | + | \mathbb{X}_{s,t} - \mathbb{X}^n_{s,t} | + | \mathbb{X}_{t,u} - \mathbb{X}^n_{t,u} |} + \underbrace{| X_{s,t} \otimes X_{t,u} - X^n_{s,t} \otimes X^n_{t,u} |}$$

for any of this terms we have :

$$| \mathbb{X} - \mathbb{X}^n | \le \| \mathbb{X} - \mathbb{X}^n \|_{2\alpha} T^{2\alpha} \longrightarrow 0$$

in coordinate : $| X^i_{s,t} X^j_{t,u} - X^{n\,i}_{s,t} X^{n\,j}_{t,u} | =$

$$= | (X^i_{s,t} - X^{n\,i}_{s,t}) X^j_{t,u} + X^{n\,i}_{s,t} (X^j_{t,u} - X^{n\,j}_{t,u}) |$$

$$\le 2 \| X - X^n \|_\alpha \| X \|_\alpha T^{2\alpha} \xrightarrow[n\to+\infty]{} 0$$

□

The natural lift of smooth paths : $X : [0,T] \longrightarrow \mathbb{R}^d$ smooth path, then

$$\mathbb{X}_{s,t} := \int_s^t (X_r - X_s) \otimes dX_r \quad \leftarrow \text{Riemann - Stieltjes sense}$$

satisfy the Chen relation. In particular $(X, \mathbb{X})$ is a rough path. $\leftarrow$ the regularity is obvious

Proof $\mathbb{X}^{i,j}_{s,t} = \int_s^t (X^i_r - X^i_s)\, dX^j_r = \int_s^u (X^i_r - X^i_s)\, dX^j_r + \int_u^t (X^i_r - X^i_s)\, dX^j_r =$

$$= \mathbb{X}^{i,j}_{s,u} + \int_u^t (X^i_r - X^i_u) + (X^i_u - X^i_s)\, dX^j_r$$

$$= \mathbb{X}^{i,j}_{s,u} + \mathbb{X}_{u,t} + X^i_{s,u} X^j_{u,t}$$

$\square$

Remarks : · From the previous proof we understand that the Chen's equality translates an idea of additivity and so it makes sense to think as the lift $\mathbb{X}$ of a rough path $\mathbf{X}$ as its "integral", postulating it as:

$$\int_s^t X_{s,r} \otimes dX_r := \mathbb{X}_{s,t}$$

· A lift $\mathbb{X}$ of a rough path $(X, \mathbb{X})$ is a "genuine" path in the sense that thanks to the Chen's equality: $\mathbb{X}_{s,t} = \mathbb{X}_{0,t} - \mathbb{X}_{0,s} - X_{0,s} \otimes X_{s,t}$ and so we reconstruct the entire lift from the path $t \longmapsto \mathbb{X}_{0,t} - \mathbb{X}_{0,s} - X_{0,s} \otimes X_{s,t}$.

The lift is not unique : $(X, \mathbb{X}) \in \mathcal{C}^\alpha$. Then $\tilde{\mathbb{X}}$ is another admissible lift of X if and only if it exists a path $F : [0,T] \longrightarrow \mathbb{R}^{d \times d}$ 2α-Hölder s.t. :

$$\tilde{\mathbb{X}}_{s,t} = \mathbb{X}_{s,t} + F_{s,t} \quad \leftarrow \text{it is } F_t - F_s$$

Proof : $\boxed{\Longrightarrow}$: we define $F_{s,t} := \tilde{\mathbb{X}}_{s,t} - \mathbb{X}_{s,t}$; from the Chen's equality we have $F_{s,t} = F_{s,u} + F_{u,t}$ $\forall s \le u \le t$, in particular $F_{0,t} = F_{0,s} + F_{s,t}$ so that $F_{s,t} = F_{0,t} - F_{0,s}$ $\forall s \le t$. Then, we can define the path $t \longmapsto F_{0,t}$ (the regularity is obvious).

$\boxed{\Longleftarrow}$: The regularity of $\tilde{\mathbb{X}}_{s,t}$. The Chen's eq.:

$$\tilde{\mathbb{X}}_{s,t} = \mathbb{X}_{s,t} + F_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t} + F_{s,u} + F_{u,t} = \tilde{\mathbb{X}}_{s,u} + \tilde{\mathbb{X}}_{u,t} + X_{s,u} \otimes X_{u,t}$$

$\square$

Remark: it is not surprising that the lift is not unique: if we think as the integral, the choice of the intermediate point changes the integral itself (Itô vs Strat. for example).

Uniform convergence and rough paths: let $\frac{1}{3} < \alpha < \beta \leq \frac{1}{2}$ and let $\mathbf{X} = (X, \mathbb{X}) \in C \times C_2$.

Let $(\mathbf{X}^n = (X^n, \mathbb{X}^n))_n \subseteq \mathscr{C}^\beta$ and assume $\sup_n |||\mathbf{X}_n|||_\beta < \infty$. Suppose further that $X^n \to X$ and $\mathbb{X}^n \to \mathbb{X}$ uniformly. Then, $\mathbf{X} \in \mathscr{C}^\beta$ and $\|\mathbf{X}^n; \mathbf{X}\|_\alpha \xrightarrow{n\to+\infty} 0$

Proof: • $\mathbf{X} \in \mathscr{C}^\beta$: • as for the paths, the lower semi continuous properties of $\|\cdot\|_\beta$ (in C^β, $C_2^{2\beta}$) imply $\|X\|_\beta, \|\mathbb{X}\|_\beta < +\infty \Rightarrow |||\mathbf{X}|||_\beta < +\infty$.

• the Chen's relation will follows from the fact that $\|\mathbf{X}^n; \mathbf{X}\|_\alpha \xrightarrow{n\to+\infty} 0$ and so it's the same as in the proof above of completeness

• $\|\mathbf{X}^n; \mathbf{X}\|_\alpha \xrightarrow{n\to+\infty} 0$: as for the paths, the interpolation prop. in C^α and $C_2^{2\alpha}$ imply the thesis.

□

Geometric rough path

Chen's relation captures the additivity of the integrals, but nothing about the integration by parts or chain rule.

↓

Recall: for a smooth path $X = (X^1, \dots, X^d) : [0,T] \to \mathbb{R}^d$ and its canonical lift $\mathbb{X}_{s,t} := \int_s^t X_{s,r} \, dX_r$,

Riemann–Stieltjes sense

we can apply the integration by parts componentwise:

$$\mathbb{X}^{i,j}_{s,t} + \mathbb{X}^{j,i}_{s,t} = \int_s^t X^i_{s,r} \, dX^j_r + \int_s^t X^j_{s,r} \, dX^i_r = X^i_{s,t} \, X^j_{s,t}$$

That becomes in a compact form: $\mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}$

So, it makes sense to define:

Weakly geometric α-Hölder rough paths: $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha$ is a weakly geometric α-Hölder rough path if $\mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}$ $\forall (s,t) \in \Delta_{[0,T]}$. We denote this space $\mathscr{C}_g^\alpha$.

$\mathcal{C}_g^\alpha$ is a complete metric space: $\mathcal{C}_g^\alpha$ is a closed subset of $\mathcal{C}^\alpha$, in particular is a complete metric space.

Proof: let $\mathbf{X}^n := (X^n, \mathbb{X}^n)$ a sequence in $\mathcal{C}_g^\alpha$ and $X \in \mathcal{C}^\alpha$ s.t.

$$|X_0 - X_0^n| + \|X - X^n\|_\alpha + \|\mathbb{X} - \mathbb{X}^n\|_{2\alpha} \xrightarrow{n\to+\infty} 0$$

We want to see that $\mathrm{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}$. So:

$$\mathrm{Sym}(\mathbb{X}_{s,t}) - \frac{1}{2} X_{s,t} \otimes X_{s,t} = \quad \mathrm{Sym}\, \mathbb{X}^n_{s,t} - X^n_{s,t} \otimes X^n_{s,t} = 0 \ \forall n$$

$$= \mathrm{Sym}(\mathbb{X}_{s,t}) - \mathrm{Sym}(\mathbb{X}^n_{s,t}) + \frac{1}{2}\left(X^n_{s,t} \otimes X^n_{s,t} - X_{s,t} \otimes X_{s,t} \right) =$$

linearity of Sym and bilinearity $\otimes$

$$= \mathrm{Sym}\left(\mathbb{X}_{s,t} - \mathbb{X}^n_{s,t} \right) + \frac{1}{2}\left((X^n - X)_{s,t} \otimes X^n_{s,t} + X_{s,t} \otimes (X^n - X)_{s,t} \right)$$

So we divide by $|t-s|^{2\alpha}$ taking the $\sup_{0\le s<t<T}$ ", using the fact that $\|\mathrm{Sym}\, A\|_{2\alpha} \le \|A\|_{2\alpha}$, we have:

$$\|\mathrm{Sym}(\mathbb{X}_{s,t}) - \frac{1}{2} X_{s,t} \otimes X_{s,t}\|_{2\alpha} \le \underbrace{\|\mathbb{X}_{s,t} - \mathbb{X}^n_{s,t}\|_{2\alpha}}_{\xrightarrow{n\to+\infty} 0} + \frac{1}{2}\left(\underbrace{\|X^n - X\|_\alpha}_{\xrightarrow{n\to+\infty}} \underbrace{\|X^n\|_\alpha}_{\substack{\text{bounded} \\ \text{as } n\to+\infty \\ \text{since } X^n \\ \text{goes to } X \\ \text{in } \|\cdot\|_\alpha}} + \underbrace{\|X\|_\alpha}_{\in \mathbb{R}} \underbrace{\|X_n - X\|_\alpha}_{\xrightarrow{n\to+\infty} 0} \right)$$

$\square$

Geometric rough paths: we define the space of geometric rough paths $\mathcal{C}_g^{0,\alpha}$ as:

$$\mathcal{C}_g^{0,\alpha} := \overline{\left\{ (X, \mathbb{X}) \in \mathcal{C}^\alpha : \underbrace{X \in C^\infty \text{ and } \mathbb{X}_{s,t} := \int_s^t X_{s,r}\, dX_r}_{\text{canonical lifting of smooth paths}} \right\}}^{\alpha\text{-Hölder rough paths distance}}$$

$\leftarrow$ by def. is a complete metric space

So, $(X, \mathbb{X}) \in \mathcal{C}_g^{0,\alpha} \iff \exists \left(X^n, \mathbb{X}^n := \int_s^t X^n_{s,r}\, dX_r \right)_n$ sequence with $X^n \in C^\infty$ s.t.

$$|X_0 - X_0^n| + \|X - X^n\|_\alpha + \|\mathbb{X} - \mathbb{X}^n\|_{2\alpha} \xrightarrow{n\to+\infty} 0.$$

Characterization of geometric rough paths: $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}_g^\alpha$. Then

$$\mathbf{X} \in \mathscr{C}_g^{0,\alpha} \iff \lim_{\delta \to 0} \sup_{|t-s|<\delta} \frac{|X_{s,t}|}{|t-s|^\alpha} = 0 \quad \text{and} \quad \lim_{\delta \to 0} \sup_{|t-s|<\delta} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} = 0$$

Proof: it's the same argument of the charach. of $C^{0,\alpha}$, that is valid also for the level 2. □

Inclusions: for $\frac{1}{3} < \alpha < \beta \le \frac{1}{2}$ we have:

$$\mathscr{C}_g^\beta \underset{\neq}{\subset} \mathscr{C}_g^{0,\alpha} \underset{\neq}{\subset} \mathscr{C}_g^\alpha \underset{\neq}{\subset} \mathscr{C}^\alpha$$

Proof: • $\mathscr{C}_g^\alpha \subseteq \mathscr{C}^\alpha$, $\mathscr{C}_g^{0,\alpha} \subseteq \mathscr{C}_g^\alpha$: obvious

• $\mathscr{C}_g^\beta \subseteq \mathscr{C}_g^{0,\alpha}$: the condition on $Sym(\mathbb{X}_{s,t})$ does not change and using the interpolation for the 1^{st} an 2^{nd} level + the previous charac. we have the thesis.

• $\mathscr{C}_g^\alpha \subsetneq \mathscr{C}^\alpha$: $X \equiv 0$, $\mathbb{X}_{s,t} = (t-s)\, Id \implies$ the Chen's equality holds so $(X, \mathbb{X}) \in \mathscr{C}^\alpha$

$\uparrow$ for $\alpha \le \frac{1}{2}$ $\in C_2^{2\alpha}$

but $Sym\, \mathbb{X}_{s,t} = \mathbb{X}_{s,t} \neq 0 = \frac{1}{2} X_{s,t} \otimes X_{s,t}$.

• $\mathscr{C}_g^{0,\alpha} \subsetneq \mathscr{C}_g^\alpha$: $X_t := t^\alpha$ $\mathbb{X}_{s,t} := \frac{1}{2}(t^\alpha - s^\alpha) \otimes (t^\alpha - s^\alpha)$, the Chen's equality hold (simple computation), it is clearly in $\mathscr{C}_g^\alpha$ but as done previously $\frac{|\varepsilon^\alpha - 0|}{|\varepsilon - 0|^\alpha} = 1 \implies \limsup_{\varepsilon \to 0} = 1 \neq 0$.

• $\mathscr{C}_g^\beta \subsetneq \mathscr{C}_g^{0,\alpha}$: $X_t := \begin{cases} t^\alpha / |\log t|, & t \in (0, e^{-1}] \\ \text{smooth} & \text{otherwise} \end{cases}$, $X_0 := 0$ and $\mathbb{X}_{s,t} := \frac{1}{2}(X_{s,t}) \otimes (X_{s,t})$

the symm rel. hold by def and:

$$\lim_{\delta \to 0} \sup_{|t-s|<\delta} \frac{|X_{s,t}^{(1)}|}{|t-s|^\alpha} \le \lim_{\delta \to 0} \frac{1}{|\log \delta|} = 0 \quad \text{and} \quad \lim_{\delta \to 0} \sup_{|t-s|<\delta} \frac{\|\mathbb{X}_{s,t}\|_{2\alpha}}{|t-s|^{2\alpha}} \le \lim_{\delta \to 0} \frac{1}{|\log \delta|} = 0$$

so $(X, \mathbb{X}) \in \mathscr{C}_g^{0,\alpha}$ but $X \in C^\beta$ so $(X, \mathbb{X}) \notin \mathscr{C}^\beta)$ □

Characterization of the second level of $\mathscr{C}_g^{0,\frac{1}{2}}$: let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}_g^{0,\frac{1}{2}}$ be a geometric rough path.

Let $\pi = \{s = u_0 < u_1 < \dots < u_N = t\}$ be a partition of $[s,t]$. Then $\mathbb{X}$ has to be equal to the Riemann-Stieltjes integra of X against itself, i.e.

$$\mathbb{X}_{s,t} = \lim_{|\pi| \to 0} \sum_{i=0}^{N-1} X_{s,u_i} \otimes X_{u_i, u_{i+1}}$$

→ Lemma : for the proof we need that if $(X, \mathbb{X})$ satisfies the Chen's relation, than

$$\mathbb{X}_{s,u_{K+1}} = \sum_{i=0}^{K} \left(\mathbb{X}_{u_i,u_{i+1}} + X_{s,u_i} \otimes X_{u_i,u_{i+1}} \right)$$

← The proof is just and induction on K

In particular for $u_{K+1} = t$:

$$\mathbb{X}_{s,t} = \sum_{i=0}^{N-1} \left(\mathbb{X}_{u_i,u_{i+1}} + X_{s,u_i} \otimes X_{u_i,u_{i+1}} \right)$$

→ Proof Using the charac. of $C_g^{\alpha,q}$ for $\alpha = \frac{1}{2}$, $\exists\, \delta$ s.t. $\sup_{|u-v|<\delta} \frac{|\mathbb{X}_{u,v}|}{|u-v|} < \varepsilon$. Then

$$\left| \mathbb{X}_{s,t} - \sum_{i=0}^{N-1} X_{s,u_i} \otimes X_{u_i,u_{i+1}} \right| \underset{\text{Lemma}}{=}$$

$$= \left| \sum_{i=0}^{N-1} \mathbb{X}_{u_i,u_{i+1}} \right| \underset{\substack{\text{we take a partition } \pi \\ \text{s.t. } |\pi| < \delta}}{<} \varepsilon \sum_{i=0}^{N-1} |u_{i+1} - u_i| = \varepsilon |t-s|.$$

Since ε is arbitrarly, the result follows.

□

Brownian motion as a rough path

Kolmogorov criterion for rough paths

<u>Recall</u>: • the norm in $L^q(\Omega, \mathcal{F}, \mathbb{P})$ is $\|X\|_{L^q} := \mathbb{E}\left[|X|^q\right]^{\frac{1}{q}}$

• $\tilde{X}$ is a modification of X if $\forall t \in [0,T]$ we have $\tilde{X}_t = X_t$ a.s.

we use $[0,T]$ as domain of $\mathbb{X}$ only because the Chen's relation holds and so it's equivalent

<u>Theorem</u>: let $(X, \mathbb{X}) : \Omega \times [0,T] \longrightarrow \mathbb{R}^d \times \mathbb{R}^{d \times d}$ a measurable stochastic process which almost surely satisfies Chen's relation. Let $q \geq 2$ and $\beta > \frac{1}{q}$ and suppose exists a constant C s.t.

$\forall s,t \in \Delta_{[0,T]}$: $\|X_{s,t}\|_{L^q} \leq C|t-s|^{\beta}$, $\|\mathbb{X}_{s,t}\|_{L^{q/2}} \leq C|t-s|^{2\beta}$.

Then, $\forall \alpha \in [0, \beta - \frac{1}{q})$ $\exists$ a modification $(\tilde{X}, \tilde{\mathbb{X}})$ of $(X, \mathbb{X})$ and random variables $K_\alpha \in L^q$, $\mathbb{K}_\alpha \in L^{q/2}$ s.t. $\forall (s,t) \in \Delta_{[0,T]}$ it holds

$$|\tilde{X}_{s,t}| \leq K_\alpha |t-s|^{\alpha} \quad \text{and} \quad |\tilde{\mathbb{X}}_{s,t}| \leq \mathbb{K}_\alpha (t-s)^{2\alpha}$$

this is obvious

In particular, if $\beta - \frac{1}{q} > \frac{1}{3}$, then for every $\alpha \in (\frac{1}{3}, \beta - \frac{1}{q})$, we have that $(\tilde{X}, \tilde{\mathbb{X}}) \in \mathcal{C}^{\alpha}$ almost surely.

<u>Proof</u>: WLOG $T=1$. The idea is to prove the inequalities for X, $\mathbb{X}$ themselves on $\bigcup_n D_n$ where $D_n := \{\frac{k}{2^n} : k = 0, 1, \dots, 2^{n-1}\}$ (the dyadic partion of $[0,1)$ with mesh size $\frac{1}{2^n}$), in this step we build K_α, $\mathbb{K}_\alpha$. Then, we use a modification of X, $\mathbb{X}$ where the inequalities hold in $[0,1]$.

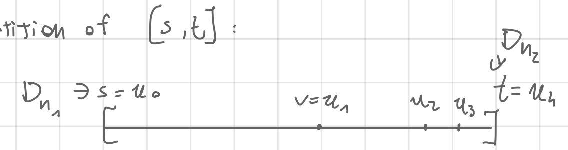
<u>Step 1</u>: • $K_n := \max_{t \in D_n} |X_{t, t+\frac{1}{2^n}}|$ $\qquad \mathbb{K}_n := \max_{t \in D_n} |\mathbb{X}_{t,t+\frac{1}{2^n}}|$. From the hip. we have

$$\mathbb{E}[K_n^q] \underset{\uparrow}{\leq} \mathbb{E}\Big[\sum_{t \in D_n} |X_{t,t+\frac{1}{2^n}}|^q\Big] \leq \sum_{t \in D_n} C^q 2^{-n\beta q} \underset{\uparrow}{=} C^q 2^{-n(\beta q - 1)}$$

$K_n \leq \sum_{t \in D_n} |X_{t,t+\frac{1}{2^n}}|$ $\qquad \# D_n = 2^n$

$$\mathbb{E}[\mathbb{K}_n^q] = \text{same with } \tfrac{q}{2} \leq C^{\frac{q}{2}} 2^{-n(\beta q - 1)}$$

• We fix $s < t$ with $s,t \in \bigcup_{n \geq 0} D_n$. Chosen $m \geq 0$ s.t. $2^{-(m+1)} < t - s \leq 2^{-m}$, we construct the following partition of $[s,t]$:



• $u_0 := s$

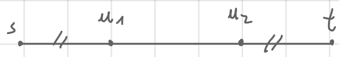
- choose the largest dyadic interval $[u_0, v]$ s.t. $[u_0, v] \subset [s,t]$ and $[u_0, v] \in D_n$ for an $n \geq m+1$ (this v must exists since $t-s > 2^{-(m+1)}$). Set $u_1 := v$.

- repeat from u_1 the previous step obtaining a partition $\{s = u_0 < u_1 < \dots < u_N = t\}$

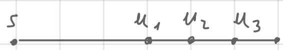
This partition has the following properties:

- $[u_i, u_{i+1}] \in D_n$ for some n s.t. $n \geq m+1$ $\forall i$ (by def.);

- for each $n \geq m+1$ there are at most two such intervals taken from D_n: if there are two intervals of D_n without s, t our algorithm selected the maximal and so it's impossibile, if there are s and t in those interval it's possible but not 3 because at least 2 are without s and t:



this is possible because $[s, u_2]$ could be not in D_n for $\forall n$



this is not possible

← more or less clear

In this way:

$$|X_{s,t}| \leq \max_{0 \leq i < N} |X_{s,u_{i+1}}| \leq \left| \sum_{i=0}^{N-1} |X_{u_i, u_{i+1}}| \right| \leq 2 \sum_{n=m+1}^{\infty} K_n \quad (*_1) \quad \text{and}$$

$$|\mathbb{X}_{s,t}| \underset{\substack{\uparrow \\ \text{Lemma: last} \\ \text{theorem previous} \\ \text{section}}}{=} \left| \sum_{i=0}^{N-1} (\mathbb{X}_{u_i, u_{i+1}} + X_{s,u_i} \otimes X_{u_i, u_{i+1}}) \right| \leq \sum_{i=0}^{N-1} (|\mathbb{X}_{u_i, u_{i+1}}| + |X_{s,u_i}| \, |X_{u_i, u_{i+1}}|)$$

$$\leq \sum_{i=0}^{N-1} |\mathbb{X}_{u_i, u_{i+1}}| + \max_{0 \leq i < N} |X_{s,u_i}| \left(\sum_{i=0}^{N-1} |X_{u_i, u_{i+1}}| \right)$$

$$\leq 2 \sum_{n=m+1}^{\infty} \mathbb{K}_n + \left(2 \sum_{n=m+1}^{\infty} K_n \right)^2 \quad (*_2)$$

- Now we define our K_α and $\mathbb{K}_\alpha$, we see that are in L^q and $L^{\frac{q}{2}}$ and that the inequalities hold in $\bigcup_n D_n$ (remember $s<t$ in $\bigcup_n D_n$):

$$K_\alpha := 2 \sum_{n=0}^{\infty} \frac{K_n}{2^{-n\alpha}}, \qquad \mathbb{K}_\alpha := 2 \sum_{n=0}^{\infty} \frac{\mathbb{K}_n}{2^{-2n\alpha}} + K_\alpha^2$$

$$\| K_\alpha \|_{L^q} \leq 2 \sum_{n=0}^{+\infty} \frac{\mathbb{E}[K_n^q]^{\frac{1}{q}}}{2^{-n\alpha}} \underset{\substack{\uparrow \\ \text{beginning} \\ \text{of Step 1}}}{\leq} 2 \sum_{n=0}^{\infty} \frac{C\, 2^{-n(\beta - \frac{1}{q})}}{2^{-n\alpha}} = 2C \sum_{n=0}^{+\infty} 2^{-n(\beta - \frac{1}{q} - \alpha)} \underset{\substack{\uparrow \\ \beta - \frac{1}{q} - \alpha > 0}}{<} \infty$$

$$\|\mathbb{K}_\alpha\|_{L^{\frac{q}{2}}} \leq \text{same passages} = 2C\sum_{n=0}^{\infty} 2^{-2n(\beta-\frac{1}{q}-\alpha)} + \|K_\alpha\|^2_{L^q} < \infty$$

$$\frac{|X_{s,t}|}{|t-s|^\alpha} \underset{(*_1)}{\leq} 2\sum_{n=m+1}^{+\infty} \frac{K_n}{2^{-(m+1)\alpha}} \underset{\substack{\uparrow \\ n \geq m+1}}{\leq} 2\sum_{n=m+1}^{+\infty}\frac{K_n}{2^{-n\alpha}} \leq 2\sum_{n=0}^{+\infty}\frac{K_n}{2^{-n\alpha}} = K_\alpha$$

$$\frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} \underset{(*_2)}{\leq} 2\sum_{n=m+1}^{+\infty}\frac{\mathbb{K}_n}{2^{-2(m+1)\alpha}} + \left(2\sum_{n=m+1}^{\infty}\frac{K_n}{2^{-(m+1)\alpha}}\right) \leq 2\sum_{n=0}^{+\infty}\frac{\mathbb{K}_n}{2^{-2n\alpha}} + K_\alpha^2 = \mathbb{K}_\alpha$$

Step 2 : we find the modification for X and $\mathbb{X}$ to extend to all $s<t$:

- let $(t_k)_{k\geq 1} \subseteq \bigcup_n D_n$ s.t. $t_k \longrightarrow t$ as $k\to+\infty$, we define:

$$\tilde{X}_t(\omega) := \lim_{k\to\infty} X_{t_k}(\omega)$$

$((X_{t_k}(\omega))_k$ is a Cauchy sequence in $\mathbb{R}^d$ since on $\bigcup_n D_n$ is α-hölder : $|X_{t_k}(\omega) - X_{t_l}(\omega)| \leq K_\alpha(\omega)|t_k-t_l|^\alpha)$

By Fatou's:

$$\|\tilde{X}_t - X_t\|_{L^\infty} \leq \liminf_k \|X_{t_k} - X_t\|_{L^q} \leq \liminf_k C|t-t_k|^\beta = 0$$

So $\tilde{X}_t = X_t$ a.s.. The inequality still holds for $\tilde{X}_t$ since if $s_k \longrightarrow s$, $t_k \to t$:

$$|\tilde{X}_{s,t}| = \lim_k |X_{s_k,t_k}| \leq \lim_k K_\alpha|t_k-s_k|^\alpha = K_\alpha|t-s|^\alpha$$

- $\forall\, 0<s\leq t<1$ (the case 0 and 1 may be dealt in the same manner) : $s_k \subset \bigcup_n D_n$ $t_k \subseteq \bigcup_n D_n$ s.t. $s_k \uparrow s$ and $t_k \downarrow t$ we define:

$$\tilde{\mathbb{X}}_{s,t}(\omega) := \lim_k \mathbb{X}_{s_k,t_k}(\omega)$$

(well defined for the same reason as before)

By Chen's equality and Fatou:

$$\|\mathbb{X}_{s,t} - \tilde{\mathbb{X}}_{s,t}\|_{L^{q/2}} \overset{\text{Fatou}}{\leq} \liminf_k \|\mathbb{X}_{s_k,t_k} - \mathbb{X}_{s,t}\|_{L^{q/2}} \overset{\text{Chen}}{=} \liminf_k \|\mathbb{X}_{s_k,s} + \mathbb{X}_{t,t_k} + X_{s_k,s}\otimes X_{s,t_k} + X_{s,t}\otimes X_{t,t_k}\|_{L^{q/2}}$$

here it is important that $s_k \leq s \leq t \leq t_k$ + usual Lemma

Hölder inequality since $\frac{1}{q}+\frac{1}{q} = \frac{1}{q/2}$

$$\leq \liminf_k \left(\|\mathbb{X}_{s_k,s}\|_{L^{q/2}} + \|\mathbb{X}_{t,t_k}\|_{L^{q/2}} + \|X_{s_k,s}\|_{L^q}\|X_{s,t_k}\|_{L^q} + \|X_{s,t}\|_{L^q}\|X_{t,t_k}\|_{L^q}\right)$$

$$\leq \liminf_k \left(C|s-s_k|^{2\beta} + C|t_k-t|^{2\beta} + C^2|s-s_k|^\beta|t_k-s|^\beta + C^2|t-s|^\beta|t_k-t|^\beta\right) = 0$$

So $\mathbb{X}_{s,t} = \tilde{\mathbb{X}}_{s,t}$ a.s. (fixed s and t). Finally the inequality:

$$|\tilde{\mathbb{X}}_{s,t}| = \lim_k |\mathbb{X}_{s_k,t_k}| \leq \lim_k K_\alpha |t_k - s_k|^{2\alpha} = K_\alpha |t-s|^{2\alpha}.$$

$\square$

Remark: when we will use this theorem we will denote the modification $(\tilde{X}, \tilde{\mathbb{X}})$ alway with $(X, \mathbb{X})$.

$\boxed{\text{Itô Brownian motion}}$

$(B, \mathbb{B}^{\text{Itô}})$: let B a d dim. BM. We define the lift

$$\mathbb{B}^{\text{Itô}}_{s,t} := \int_s^t B_{s,r} \otimes dB_r \qquad \forall (s,t) \in \Delta_{[0,T]}.$$

understood in the Itô sense

$(B, \mathbb{B}^{\text{Itô}})$ is a rough path: $\forall \alpha \in \left(\frac{1}{3}; \frac{1}{2}\right)$ we have

$$\mathbf{B} := (B, \mathbb{B}^{\text{Itô}}) \in \mathscr{C}^\alpha([0,T]; \mathbb{R}^d) \quad \text{a.s.}$$

Proof: • Chen's equality: since the Itô integral is additive, it's the exact proof as for smooth paths.

• Regularity: we want to use the Kolmogorov criterion (we find a modification but it's ok since a modification does not change the law of the stoch. process and the modification we find is α-Hölder continuous so it is still a BM): let $q \geq 2$

$$\| B_{s,t} \|_{L^q} = \| (t-s)^{\frac{1}{2}} B_1 \|_{L^q} = (t-s)^{\frac{1}{2}} \| B_1 \|_{L^q}$$

$B_{s,t} \sim (t-s)^{\frac{1}{2}} B_1$ and to compute an expectation we only need the law

$$\| \mathbb{B}^{\text{Itô}}_{s,t} \|_{L^{\frac{q}{2}}} = \mathbb{E}\left[\left| \int_s^t B_{s,r} \otimes dB_r \right|^{\frac{q}{2}} \right]^{\frac{2}{q}} \leq$$

Burkhölder-Davis-Gundy: for a local martingale M_t with $M_0 = 0$ we have for $p \geq 1$ $\exists C_p$:

$$\mathbb{E}\left[\sup_{0 \leq u \leq t} |M_u|^p\right] \leq C_p \mathbb{E}\left[\langle M \rangle_t^{p/2}\right]$$

But $M_u := \int_s^u B_{s,r} \otimes dB_r$ is a continuous martingale and $\langle M \rangle_t = \int_s^t |B_{s,r}|^2 dr$ (is the trace of the covariation, it's unidim.)

$$\leq C_q^{\frac{2}{q}} \mathbb{E}\left[\left(\int_s^t |B_{s,r}|^2 dr \right)^{\frac{q}{4}} \right]^{\frac{2}{q}}$$

$\int_s^t |B_{s,r}|^2 dr \leq \sup_r |B_{s,r}|^2 \cdot |t-s|$

$$\leq C_q^{\frac{2}{q}} \mathbb{E}\left[\sup_{r \in [s,t]} |B_{s,r}|^{\frac{q}{2}} \right]^{\frac{2}{q}} |t-s|^{\frac{1}{2}}$$

BDG: $\langle B_{s,t} \rangle = d(t-s)$

$$\leq C_q^{\frac{2}{q}} C_q^{\frac{1}{q}} d^{\frac{1}{2}} |t-s|^{\frac{1}{2}} |t-s|^{\frac{1}{2}} = K |t-s|$$

So we can apply Kolmogorov with $\beta = \frac{1}{2}$ and $q \geq 2$, hence $(B, \mathbb{B}^{Itô}) \in \mathcal{C}^{\alpha}$ $\forall \alpha \in \left(\frac{1}{3}, \frac{1}{2} - \frac{1}{q}\right)$,

taking $q \to +\infty$ we have the th. $\square$

The Itô rough path is not weakly geometric: for $(B, \mathbb{B}^{Itô})$ holds:

$$\mathrm{Sym}\left(\mathbb{B}^{Itô}_{s,t}\right) = \frac{1}{2}\left(B_{s,t} \otimes B_{s,t} - (t-s)\, Id\right)$$

Proof: $\mathrm{Sym}\left(\mathbb{B}^{Itô}_{s,t}\right)^{i,j} = \dfrac{\left(\mathbb{B}^{Itô}_{s,t}\right)^{i,j} + \left(\mathbb{B}^{Itô}_{s,t}\right)^{j,i}}{2} = \dfrac{\int_s^t B^i_{s,r}\, dB^j_r + \int_s^t B^j_{s,r}\, dB^i_r}{2} =$

$$= \frac{1}{2}\left(B_{s,t} \otimes B_{s,t} - (t-s)\, Id\right)$$

using Itô formula on $X_r := B^i_{s,r}$, $Y_r = B^j_{s,r}$ we have $X_s = Y_s = 0$ and $dX_r = dB^i_r$, $dY_r = dB^j_r$

and $[X,Y]_{s,t} = [B^i, B^j]_{s,t} = \delta_{ij}(t-s)$: $X_t Y_t = X_s Y_s + \int_s^t X_r\, dY_r + \int_s^t Y_r\, dX_r + [X,Y]_{s,t}$

$\square$

Stratonovich Brownian motion

Recall: for 1-dim. continuous semimart. X, Y, the Stratonovich integral of Y against X is

$$\int_0^t Y_s \circ dX_s := \int_0^t Y_s\, dX_s + \frac{1}{2}\langle X, Y\rangle_t$$ ← quadratic covariation ($[\cdot,\cdot]$ equivalent)

(it is defined as $\lim\limits_{n\to+\infty} \sum\limits_{i=0}^{N-1} \frac{1}{2}\left(Y_{t_i^n} - Y_{t_{i+1}^n}\right)\left(X_{t_{i+1}^n} - X_{t_i^n}\right)$ (the average between the endpoints

of $[t_i, t_{i+1}]$ and not only X_{t_i} as in Itô). In particular $\int_0^t Y_s \circ dX_s$ in not a martingale in

general but:

- $X_t Y_t = X_0 Y_0 + \int_0^t X_s \circ dY_s + \int_0^t Y_s \circ dX_s$ (Integr. by parts)

- $f(X_t) = f(X_0) + \int_0^t Df(X_s) \circ dX_s$ (Chain rule)

$(B, \mathbb{B}^{strat})$: let B a d-dim. BM. We define the lift

$$\mathbb{B}^{strat} := \int_s^t B_{s,r} \circ \otimes\, dB_r \qquad (s,t) \in \Delta_{[0,T]}$$

Now we have immediately $\mathrm{Sym}\left(\mathbb{B}^{strat}\right) = \mathrm{Sym}\left(\mathbb{B}^{Itô}_{s,t}\right) + \frac{1}{2}(t-s)\, Id = \frac{1}{2} B_{s,t} \otimes B_{s,t}$, so if $(B, \mathbb{B}^{strat}) \in \mathcal{C}^{\alpha}$ then is in $\mathcal{C}^{\alpha}_g$. But we can do better:

$(B, \mathbb{B}^{Strat})$ is a geometric rough path : $\forall \alpha \in \left(\frac{1}{3}, \frac{1}{2}\right)$ we have that

$$(B, \mathbb{B}^{Strat}) \in \mathcal{C}^{0,\alpha}_g([0,T]; \mathbb{R}^d) \quad \text{a.s.}$$

Proof Chen's equality : by additivity and using $\langle 1, X_r \rangle = 1$ so that as for the Itô integral:

$$\forall s \le u \le t \quad \int_u^t B^i_{s,u} \circ dB^j_r = \int_u^t B^i_{s,u}\, dB^j_r + \underbrace{\langle B^i_{s,u}, B^j_r \rangle_{u,t}}_{=0,\ B^i_{s,u} \text{ constant in } [u,t]} = B^i_{s,u} B^j_{u,t}$$

Regularity : B is in C^{α} for the Itô-case (we always take the right modification).

and $\mathbb{B}^{Strat} = \mathbb{B}^{Ito} + \frac{1}{2}(t-s)\, Id$ but $\mathbb{B}^{Ito} \in \mathcal{C}^{2\alpha}_2$ and $(s,t) \mapsto \frac{1}{2}(t-s)\, Id$ is 1-Hölder continuous

and so $\mathbb{B}^{Strat} \in C^{2\alpha}_2$

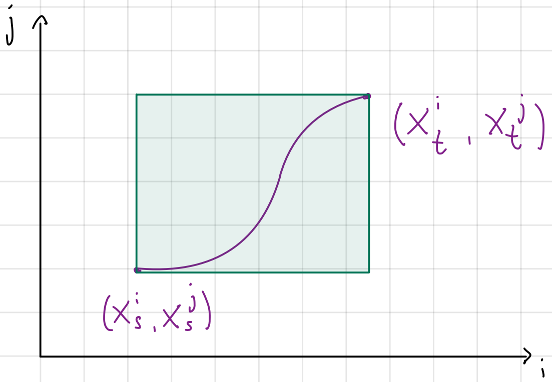
Geometric : since $(B, \mathbb{B}^{Strat}) \in \mathcal{C}^{\beta}_g \ \forall \beta \in \left(\frac{1}{3}, \frac{1}{2}\right)$, given $\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right)$ $\exists \beta$ s.t. $\beta \in \left(\alpha, \frac{1}{2}\right)$

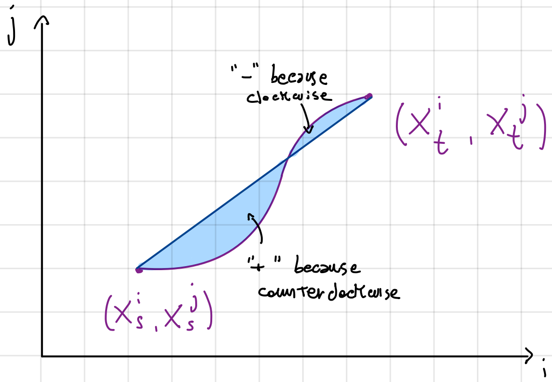
and by $\mathcal{C}^{\beta}_g \subset \mathcal{C}^{0,\alpha}_g$ we have the result. $\square$

Remarks : • For a weakly geom. rough path, the new information encoded by $\mathbb{X}$ is all in the antisimmetric part $\text{Anti}(\mathbb{X}_{s,t})^{ij} = \frac{\mathbb{X}^{ij}_{s,t} - \mathbb{X}^{ji}_{s,t}}{2}$ since $\text{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2}(X_{s,t} \otimes X_{s,t})$

and so knowing only the path X we know $\text{Sym}(\mathbb{X}_{s,t})$ as well.

For a smooth path with its canonical lift it is clear the interpretation: (for a geometric one is the approximation of the areas)





Green area $= X^i_{s,t} X^j_{s,t} = 2\, \text{Sym}^{i,j}(\mathbb{X})$

We don't see how the components interact with each other

Blue area $= \frac{1}{2}\left(\int_s^t X^i_{s,r}\, dX^j_r + \int_s^t X^j_{s,r}\, dX^i_r\right)$

$= \text{Anti}(\mathbb{X}_{s,t})^{ij}$

We have the interaction between X^i_t and X^j_t

- In the particular case of the Itô rough path it is true what we have just say (although is not a weakly geom. rough path) because $\mathrm{Sym}\, \mathbb{B}^{\text{Itô}}_{s,t} = \frac{1}{2}\left(B_{s,t} \otimes B_{s,t} - (t-s)\, I_d\right)$.
- Since 1×1 - matrix is already symm. any 1-dim $X \in C^{\alpha}$ may be lifted to a weakly geometric rough path simply seting $\mathbb{X}_{s,t} := \frac{1}{2}(X_{s,t})^2$

Integration

The sewing lemma

Theorem (the sewing lemma): let $(E, \|\cdot\|)$ be a Banach space and let $A : \Delta_{[0,T]} \longrightarrow E$ a continuous function. For each triplet $0 \leq s \leq u \leq t \leq T$ we set:

$$\delta A_{s,u,t} := A_{s,t} - A_{s,u} - A_{u,t}.$$

Suppose there exists a constant $\lambda \geq 0$ and $\varepsilon > 0$ s.t.

$$\|\delta A_{s,u,t}\| \leq \lambda |t-s|^{1+\varepsilon} \qquad \forall\ s \leq u \leq t \leq T.$$

Then, $\exists\ \gamma : [0,T] \longrightarrow E$ continuous with $\gamma_0 = 0$ s.t.

$$\|\gamma_t - \gamma_s - A_{s,t}\| \leq C \lambda |t-s|^{1+\varepsilon} \qquad \forall (s,t) \in \Delta_{[0,T]} \text{ and } C \text{ depends only on } \varepsilon.$$

Moreover, $\forall (s,t) \in \Delta_{[0,T]}$ we have that

$$\lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} A_{u,v} = \gamma_t - \gamma_s . \qquad (\pi \text{ partition of } [s,t])$$

→ Comment: Why "sewing"? Suppose A is additive in the sense that $\delta A_{s,u,t} = 0 \quad \forall s < u < t$ $(A_{s,t} = A_{s,u} + A_{u,t})$, then we can construct γ simply setting $\gamma_t := A_{0,t}$; infact $A_{0,0} = 0$ $\left(\|\delta A_{t,t,t}\| \leq \lambda\, 0^{1+\varepsilon} \Rightarrow A_{t,t} \equiv 0 \right)$, $\gamma_t - \gamma_s - A_{s,t} = A_{0,t} - A_{0,s} - A_{s,t} = 0$ by additivity and $\lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} A_{u,v} = \lim_{|\pi| \to 0} A_{s,t} = A_{s,t} = A_{0,t} - A_{0,s} = \gamma_t - \gamma_s$.

This means that we can sew a two index map $A_{s,t}$ in a path γ_t in a trivial way. The sewing lemma tells us that this is still possible if the "error of additivity" is small enough.

Proof: the idea is to find a real additive 2-parameter function $\Gamma_{s,t}$, so that, as the "Comment" suggestes, we can define $\gamma_t = \Gamma_{0,t}$ and see that $A_{s,t}$ is not "too far" from $\Gamma_{s,t}$.

Step 1: Construct a sequence $(A^n_{s,t})_n$ (approximating $A_{s,t}$) using dyadic partition, see that the sequence is Cauchy and hence has limit in E.

- Fix $(s,t) \in \Delta_{[0,T]}$ and $\forall n \geq 0$ let $\{ s = t_0^n < t_1^n < \dots < t_{2^n}^n = t \}$ the dyadic partition of $[s,t]$ $\left(t_i^n = s + \frac{i}{2^n}(t-s) \right)$ which mesh size is $|\pi^n| = \frac{|t-s|}{2^n}$. We define

$$A^n_{s,t} := \sum_{i=0}^{2^n-1} A_{t_i^n t_{i+1}^n}$$

- Calling u_i^n the midpoint of $[t_i^n, t_{i+1}^n]$, basically $[t_i^n, u_i^n]$ and $[u_i^n, t_{i+1}^n]$ are in the partition of the $(n+1)$-step so:

$$A^n_{s,t} - A^{n+1}_{s,t} = \sum_{i=0}^{2^n-1} A_{t_i^n t_{i+1}^n} - \sum_{i=0}^{2^{n+1}-1} A_{t_i^{n+1} t_{i+1}^{n+1}}$$

$$= \sum_{i=0}^{2^n-1} A_{t_i^n t_{i+1}^n} - \sum_{i=0}^{2^n-1} \left(A_{t_i^n u_i^n} + A_{u_i^n t_{i+1}^n} \right)$$

$$= \sum_{i=0}^{2^n-1} \delta A_{t_i^n u_i^n t_{i+1}^n} \quad . \quad \text{So:}$$

$$\| A^n_{s,t} - A^{n+1}_{s,t} \| \leq \sum_{i=0}^{2^n-1} \| \delta A_{t_i^n u_i^n t_{i+1}^n} \| \leq \sum_{i=0}^{2^n-1} \lambda \left(t_{i+1}^n - t_i^n \right)^{1+\varepsilon} \overset{\text{dyadic part. def.}}{=}$$

$$= \lambda |t-s|^{1+\varepsilon} \sum_{i=0}^{2^n-1} 2^{-n(1+\varepsilon)} \underset{\sum_{i=0}^{2^n-1} 1 = 2^n}{=} \lambda |t-s|^{1+\varepsilon} 2^{-n\varepsilon}$$

- The sequence is Cauchy: m_1, m_2 (WLOG $m_1 < m_2$) then

$$\| A^{m_1}_{s,t} - A^{m_2}_{s,t} \| = \| \sum_{n=m_1}^{m_2-1} A^n_{s,t} - A^{n+1}_{s,t} \| \leq \sum_{n=m_1}^{m_2-1} \| A^n_{s,t} - A^{n+1}_{s,t} \|$$

$$\leq \sum_{n=m_1}^{+\infty} \| A^n_{s,t} - A^{n+1}_{s,t} \| \leq \lambda |t-s|^{1+\varepsilon} \sum_{n=m_1}^{+\infty} 2^{-n\varepsilon} = \lambda |t-s|^{1+\varepsilon} \frac{2^{-m_1\varepsilon}}{1-2^{-m_1\varepsilon}}$$

So as $m_1 \to +\infty$, $\| A^{m_1}_{s,t} - A^{m_2}_{s,t} \| \xrightarrow{m_1 \to \infty} 0$ and so is Cauchy.

- Since is Banach we can set:

$$\Gamma_{s,t} := \lim_n A^n_{s,t}$$

<u>Step 2</u>: setting $\gamma_t := \Gamma_{0,t}$ we obtain the th.:

- γ_t is continuous: $\| \Gamma_{s,t} - A^n_{s,t} \| = \| \lim_m \left(A^m_{s,t} - A^n_{s,t} \right) \| =$

$$= \| \lim_m \sum_{k=n}^{m-1} A^{k+1}_{s,t} - A^k_{s,t} \| = \| \sum_{k=n}^{+\infty} A^{k+1}_{s,t} - A^k_{s,t} \| \leq \sum_{k=n}^{\infty} \| A^k_{s,t} - A^{k+1}_{s,t} \| \leq \lambda |t-s|^{1+\varepsilon} \frac{2^{-n\varepsilon}}{1-2^{-\varepsilon}}$$

So the convergence is uniform and since A^n is continuous $\forall n$, Γ is also continuous.

in the sense $(s,t) \mapsto A^n_{s,t}$
(not difficult to show)

- $\gamma_0 = 0$: $\quad \gamma_0 = \lim_n A^n_{0,0} \underset{\substack{\uparrow \\ \text{see "Comment"}}}{=} 0$

- Bound on $\| \gamma_t - \gamma_s - A_{s,t} \|$: $\quad \Gamma$ is additive infact if u is a dyadic time in $[s,t]$

the additive follows from def:

$$\Gamma_{s,t} = \lim_n A^n_{s,t} = \lim_n \sum_{i=0}^{2^n-1} A_{t_i^n, t_{i+1}^n} \underset{\substack{\uparrow \\ u = t_j \text{ for a } j \in \{0,\dots,2^n-1\}}}{=} \lim_n \left(\sum_{i=0}^{j-1} A_{t_i^n, t_{i+1}^n} + \sum_{i=j}^{2^n-1} A_{t_i^n, t_{i+1}^n} \right) =$$

$$= \Gamma_{s,u} + \Gamma_{u,t}$$

but Γ is continuous so $\forall u \in [s,t]$ we can take $u_k \xrightarrow{k} u$ with u_k dyadic for $[s,t]$

so that:

$$\Gamma_{s,u} + \Gamma_{u,t} \underset{\substack{\uparrow \\ \Gamma \text{ continuous}}}{=} \lim_k \Gamma_{s,u_k} + \Gamma_{u_k,t} = \lim_k \Gamma_{s,t} = \Gamma_{s,t}$$

So we have $\gamma_t - \gamma_s = \Gamma_{0,t} - \Gamma_{0,s} \underset{\substack{\uparrow \\ \text{additive}}}{=} \Gamma_{s,t}$ and so

$$\| \gamma_t - \gamma_s - A_{s,t} \| = \| \Gamma_{s,t} - A_{s,t} \| \leq \underbrace{\| \Gamma_{s,t} - A^n_{s,t} \|}_{\leq \lambda |t-s|^{1+\varepsilon} \frac{2^{-k\varepsilon}}{1-2^{-\varepsilon}}} + \underbrace{\| A^n_{s,t} - A_{s,t} \|}_{\substack{A_{s,t} = A^0_{s,t} \text{ so} \\ A^n_{s,t} - A_{s,t} = \sum_{k=0}^{n-1} A^{k+1}_{s,t} - A^k_{s,t}}}$$

$$\leq \lambda |t-s|^{1+\varepsilon} \frac{2^{-k\varepsilon}}{1-2^{\varepsilon}} + \sum_{k=0}^{n-1} \underbrace{\| A^{k+1}_{s,t} - A^k_{s,t} \|}_{\leq \lambda |t-s|^{1+\varepsilon} 2^{-k\varepsilon}} \quad \text{and} \quad \sum_{k=0}^{n-1} 2^{-k\varepsilon} \leq \sum_{k=0}^{+\infty} 2^{-k\varepsilon} = \frac{1}{1-2^{-\varepsilon}}$$

$$\leq \lambda |t-s|^{1+\varepsilon} \frac{2^{-k\varepsilon}}{1-2^{\varepsilon}} + \frac{\lambda |t-s|^{1+\varepsilon}}{1-2^{-\varepsilon}} \xrightarrow{k \to +\infty} \leq \frac{\lambda |t-s|^{1+\varepsilon}}{1-2^{-\varepsilon}} = C_\varepsilon \lambda |t-s|^{1+\varepsilon}$$

- Bound on $\| \gamma_t - \gamma_s - \sum_{i=0}^{N-1} A_{t_i, t_{i+1}} \|$:

$$\| \gamma_t - \gamma_s - \sum_{i=0}^{N-1} A_{t_i, t_{i+1}} \| \underset{\substack{\uparrow \\ \pi = \{ s = t_0 < t_1 < \dots < t_N = t \}}}{=} \| \sum_{i=0}^{N-1} \gamma_{t_{i+1}} - \gamma_{t_i} - A_{t_i, t_{i+1}} \| \leq$$

$$\leq \sum_{i=0}^{N-1} \| \gamma_{t_{i+1}} - \gamma_{t_i} - A_{t_i, t_{i+1}} \| \underset{\substack{\uparrow \\ \text{previous bound}}}{\leq} \frac{\lambda}{1-2^{-\varepsilon}} \sum_{i=0}^{N-1} | t_{i+1} - t_i |^{1-\varepsilon} =$$

$$\leq \frac{\lambda}{1-2^{-\varepsilon}} |t-s| \, |\pi|^{\varepsilon} \xrightarrow{|\pi| \to 0} 0 .$$

$\square$

Young Integration

<u>Young Integral</u>: $X \in C^{\alpha}([0,T]; E)$, $Y \in C^{\beta}([0,T]; \mathcal{L}(E, \mathcal{G}))$ for $\alpha, \beta \in (0,1]$ with $\alpha+\beta > 1$. Then

$\to E, \mathcal{G}$ Banach (finite dim.)

$$\int_s^t Y_u \, dX_u := \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_u X_{u,v}$$

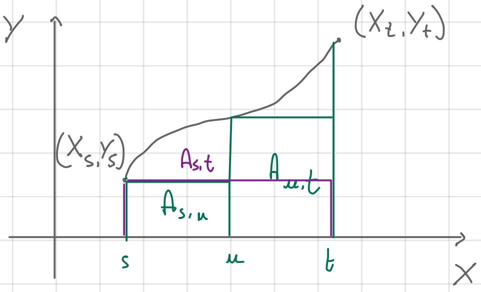
π partition of $[s,t]$

exists $\forall (s,t) \in \Delta_{[0,T]}$. This is called the Young integral of Y against X. Moreover we have the estimation

$$\left| \int_s^t Y_u \, dX_u - Y_s X_{s,t} \right| \leq C \, \|Y\|_\beta \, \|X\|_\alpha \, |t-s|^{\alpha+\beta}$$

$\forall (s,t) \in \Delta_{[0,T]}$ where C depends only on $\alpha+\beta$.

$\to$ <u>Comment</u>: the idea using the sewing lemma is clear $(d=m=1)$:



$(s,t) \longmapsto A_{s,t} := Y_s X_{s,t}$, it is not additive as we see in the picture on the left but if the error is low enough we can sew together the pieces in a path (the integral) s.t. is proper the limit of the sum of these areas.

<u>Proof</u>: <u>Existence</u>: we define $A_{s,t} := Y_s X_{s,t}$. Then A is clearly continuous. Now

$$|\delta A_{s,u,t}| = |A_{s,t} - A_{s,u} - A_{u,t}| = |Y_s X_{s,t} - Y_s X_{s,u} - Y_u X_{u,t}|$$

$$= |Y_s X_t - \cancel{Y_s X_s} - Y_s X_u + \cancel{Y_s X_s} - Y_u X_t + Y_u X_u|$$

$$= |Y_s X_{u,t} - Y_u X_{u,t}| = |Y_{u,s} X_{u,t}|$$

$$\leq \|Y\|_\beta \, \|X\|_\alpha \, |t-s|^{\alpha+\beta}$$

So applying the sewing lemma we have finished setting $\int_s^t Y_u \, dX_u := \gamma_t - \gamma_s$

<u>Bound</u>: by the sewing lemma:

$$\left| \int_s^t Y_u dX_u - Y_s X_{s,t} \right| \leq \underset{\text{depending on } \alpha+\beta}{C} \, |t-s|^{\alpha+\beta} \, \|Y\|_\beta \, \|X\|_\alpha$$

$\square$

The Young integral does not depend on the point $r \in [u,v]$: $X \in C^\alpha$, $Y \in C^\beta$ for some $\alpha, \beta \in (0,1]$ with $\alpha+\beta>1$. For a partition π and an interval $[u,v] \in \pi$, let $r \in [u,v]$ an arbitrary point in $[u,v]$. Then we have:

$$\int_0^t Y_u \, dX_u = \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_r X_{u,v}$$

Proof

$$\lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_r X_{u,v} = \lim_{|\pi| \to 0} \left(\sum_{[u,v] \in \pi} Y_u X_{u,v} + \sum_{[u,v] \in \pi} Y_{u,r} X_{u,v} \right) =$$

$$\left| \sum_{[u,v] \in \pi} Y_{u,r} X_{u,v} \right| \leq \qquad \nearrow \quad = \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_u X_{u,v} + 0$$

$$\sum_{[u,v] \in \pi} |Y_{u,r} X_{u,v}| \leq \qquad = \int_s^t Y_u \, dX_u .$$

$$\|Y\|_\alpha \|X\|_\beta \sum_{[u,v] \in \pi} |u-v|^{\alpha+\beta} \leq$$

$$|u-v|^{\alpha+\beta} = \underbrace{|u-v|^{\alpha+\beta-1}}_{\leq |\pi|} |u-v|$$

$$\|Y\|_\alpha \|X\|_\beta \, T \, |\pi|^{\alpha+\beta-1} \xrightarrow{|\pi| \to 0} 0$$

$\alpha+\beta-1>1$

$\square$

The Young integral is locally lip. : the Young integral is a well defined locally lipschitz map ($\alpha, \beta \in (0,1]$ with $\alpha+\beta>1$)

we use the norm $|X_0| + \|X\|_\alpha + |Y_0| + \|Y\|_\beta$

$$C^\alpha \times C^\beta \longrightarrow C^\alpha$$

$$(X, Y) \longmapsto \left(t \mapsto \int_0^t Y_u \, dX_u \right)$$

Indeed it holds

$$\underbrace{\left\| \int_0^\cdot Y_u dX_u - \int_0^\cdot \tilde{Y}_u d\tilde{X}_u \right\|_\alpha}_{} \leq C \underbrace{\left(\Big(\underbrace{|Y_0 - \tilde{Y}_0| + \|Y - \tilde{Y}\|_\beta}_{= |Y-\tilde{Y}|_\beta}\Big) \|X\|_\alpha + \Big(\underbrace{|\tilde{Y}_0| + \|\tilde{Y}\|_\beta}_{|\tilde{Y}|_\beta}\Big) \|X - \tilde{X}\|_\alpha \right)}_{}$$

this is actually the norm in C^α because at $\int_0^0 \cdot = 0$

in a ball of radius R large enough $\|X\|_\alpha, |\tilde{Y}_0|, \|\tilde{Y}\|_\beta \leq R$

$$= CR \left(|Y_0 - \tilde{Y}_0| + \|Y - \tilde{Y}\|_\beta + \|X - \tilde{X}\|_\alpha \right)$$

← we have been "very large" since there is also $|X_0 - \tilde{X}_0|$; in general does not affect since $\tilde{X} = X + c$ gives the same integ.

so $\| \Phi(X,Y) - \Phi(\tilde{X}, \tilde{Y}) \|_{C^\alpha} \leq C_R \, \| (X - \tilde{X}, Y - \tilde{Y}) \|_{C^\alpha \times C^\beta}$

the real norm

(def. of locally lip.)

$\hookrightarrow$ Proof: We want to apply the sewing lemma so:

$$A_{s,t} := Y_s X_{s,t} \ , \ \tilde{A}_{s,t} = \tilde{Y}_s \tilde{X}_{s,t} \ , \quad \Delta := A - \tilde{A} .$$

Then:

$$|\delta \Delta_{s,u,t}| = |\Delta_{s,t} - \Delta_{s,u} - \Delta_{u,t}| = | A_{s,t} - \tilde{A}_{s,t} - A_{s,u} + \tilde{A}_{s,u} - A_{u,t} + \tilde{A}_{s,u}|$$

$$= | Y_s X_{s,t} - \tilde{Y}_s X_{s,t} - Y_s X_{s,u} + \tilde{Y}_s \tilde{X}_{s,u} - Y_u X_{u,t} + \tilde{Y}_u \tilde{X}_{u,t}|$$

$$= | Y_{s,u} X_{u,t} - \tilde{Y}_{s,u} \tilde{X}_{u,t} | \leq \quad - \tilde{Y}_{s,u} X_{u,t} + \tilde{Y}_{s,u} X_{u,t}$$

$$\leq | Y_{s,u} - \tilde{Y}_{s,u}| |X_{u,t}| + |X_{u,t} - \tilde{X}_{u,t}| |\tilde{Y}_{s,u}|$$

$$\leq \left(\| Y - Y\|_\beta \|X\|_\alpha + \| X - \tilde{X}\|_\alpha \|\tilde{Y}\|_\beta\right) (t-s)^{\alpha+\beta}$$

So by the sewing lemma we have a path γ_t and a constant C s.t.

$$|\gamma_t - \gamma_s - \Delta_{s,t}| \leq C \left(\| Y - Y\|_\beta \|X\|_\alpha + \| X - \tilde{X}\|_\alpha \|\tilde{Y}\|_\beta\right) (t-s)^{\alpha+\beta} \quad \text{so}$$

$$\frac{|\gamma_t - \gamma_s|}{(t-s)^\alpha} \leq C \left(\| Y - \tilde{Y}\|_\beta \|X\|_\alpha + \| X - \tilde{X}\|_\alpha \|\tilde{Y}\|_\beta\right) (t-s)^{\beta} + \frac{|\Delta_{s,t}|}{(t-s)^\alpha} \qquad (*_1)$$

Now the LHS of $(*_1)$ has $\gamma_t - \gamma_s \overset{\text{sewing lemma}}{=} \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} \Delta_{u,v} = \lim_{|\pi| \to 0} \sum_{u,v \in \pi} \left(A_{u,v} - \sum_{u,v} \tilde{A}_{u,v}\right) =$

$$= \int_s^t Y_u dX_u - \int_s^t \tilde{Y}_u d\tilde{X}_u \qquad (*_2)$$

In the RHS of $(*_1)$ there is $|\Delta_{s,t}| = | Y_s X_{s,t} - \tilde{Y}_s \tilde{X}_{s,t}| \leq \quad - \tilde{Y}_s X_{s,t} + \tilde{Y}_s X_{s,t}$

$$\leq | Y_s - \tilde{Y}_s| |X_{s,t}| + |X_{s,t} - \tilde{X}_{s,t}| |\tilde{Y}_s|$$

in general $\|K\|_\infty \leq |K_0| + T^\beta \|K\|_\beta$

$$\leq \left(\| Y - \tilde{Y}\|_\infty \|X_{s,t}\|_\alpha + \| X - \tilde{X}\|_\alpha \|\tilde{Y}\|_\infty\right) |t-s|^\alpha$$

$$(*_3) \qquad \leq \left(\left(| Y_0 - \tilde{Y}_0| + T^\beta \| Y - \tilde{Y}\|_\beta\right) \|X\|_\alpha + \|X - \tilde{X}\|_\alpha \left(|\tilde{Y}_0| + T^\beta \|\tilde{Y}\|_\beta\right)\right) |t-s|^\alpha$$

Putting $(*_2)$ and $(*_3)$ in $(*_1)$, using $(t-s)^\beta \leq T^\beta$ and taking the $\sup_{s<t}$:

$$\left\| \int_s^t Y_u dX_u - \int_s^t \tilde{Y}_u d\tilde{X}_u \right\|_\alpha \leq C \left(\| Y - \tilde{Y}\|_\beta \|X\|_\alpha + \| X - \tilde{X}\|_\alpha \|\tilde{Y}\|_\beta\right) T^\beta +$$

$$+ \left(| Y_0 - \tilde{Y}_0| + T^\beta \| Y - \tilde{Y}\|_\beta\right) \|X\|_\alpha + \|X - \tilde{X}\|_\alpha \left(|\tilde{Y}_0| + T^\beta \|\tilde{Y}\|_\beta\right)$$

$$= \|X\|_\alpha \Big(\underbrace{C T^\beta \|Y-\tilde Y\|_\beta + |Y_0-\tilde Y_0| + T^\beta \|Y-\tilde Y\|_\beta}_{} \Big) + \|X-\tilde X\|_\alpha \Big(\underbrace{C T^\beta \|\tilde Y\|_\beta + |\tilde Y_0| + T^\beta \|\tilde Y\|_\beta}_{} \Big)$$

$$= (1+C)\left(T^\beta \|Y-\tilde Y\|_\beta\right) + |Y_0-\tilde Y_0| \qquad = (1+C) T^\beta \|\tilde Y\|_\beta + |\tilde Y_0|$$

$$\le (1+C)\left(T^\beta \|Y-\tilde Y\|_\beta + |Y_0-\tilde Y_0|\right) \qquad \le (1+C)\left(T^\beta \|\tilde Y\|_\beta + |\tilde Y_0|\right)$$

$T^\beta A + B \le$
$\le (1+T^\beta)(A+B)$

$$\le (1+C)(1+T^\beta)\left(\|Y-\tilde Y\|_\beta + |Y_0-\tilde Y_0|\right) \qquad \le (1+C)(1+T^\beta)\left(\|\tilde Y\|_\beta + |\tilde Y_0|\right)$$

$$\le (1+C)(1+T^\beta)\Big(\|X\|_\alpha \left(\|Y-\tilde Y\|_\beta + |Y_0-\tilde Y_0|\right) + \|X-\tilde X\|_\alpha \left(\|\tilde Y\|_\beta + |\tilde Y_0|\right)\Big)$$

$\square$

Young Integration satisfies the integration by parts: let $X \in C^\alpha([0,T],\mathbb{R}^d)$, $Y \in C^\beta([0,T],\mathbb{R}^m)$ with $\alpha+\beta>1$ $(\alpha,\beta \in (0,1])$.

Then: $B(X_T,Y_T) = B(X_0,Y_0) + \int_0^T B(X_u, dY_u) + \int_0^T B(dX_u, Y_u)$ $\forall t \in [0,T]$

well defined since $B(X_u,\cdot) \in \mathcal{L}(\mathbb{R}^m,\mathbb{R}^n)$ and $B(X_t,\cdot) - B(X_s,\cdot) \le \|B\|_{op} |X_{s,t}| \lesssim |t-s|^\alpha$ (same for $B(\cdot, Y_u)$)

for every $B \in Bil(\mathbb{R}^d \times \mathbb{R}^m ; \mathbb{R}^n)$. For example $B(v,w) = v \cdot w$ or $B(v,w) = v \otimes w$

Proof: WLOG $t=T$.

$$B(X_T,Y_T) - B(X_0,Y_0) = \sum_{[s,t]\in\pi} \left(B(X_t,Y_t) - B(X_s,Y_s)\right) = \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} \left(B(X_t,Y_t) - B(X_s,Y_s)\right)$$

telescopic

it is constant w.r.t. $|\pi|$

$$= \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} \Big(B(X_{s,t},Y_s) + B(X_s,Y_{s,t}) - B(X_{s,t},Y_s) - B(X_s,Y_{s,t}) + B(X_t,Y_t) - B(X_s,Y_s) \Big)$$

I want them, so I put them and subtract

bilinearity of B

$$= \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} B(X_{s,t},Y_s) + B(X_s,Y_{s,t}) + B(X_{s,t},Y_{s,t}) =$$

$$\lim_{|\pi|\to 0} \Big| \sum_{[s,t]\in\pi} B(X_{s,t},Y_{s,t}) \Big| \le$$

$|B(X_{s,t},Y_{s,t})| \le \|B\|_{op} |X_{s,t}| |Y_{s,t}|$

$$= \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} B(X_{s,t},Y_s) + B(X_s,Y_{s,t}) \qquad \le \lim_{|\pi|\to 0} \|B\|_{op} \|X\|_\alpha \|Y\|_\beta \sum_{[s,t]\in\pi} |t-s|^{\alpha+\beta}$$

$$= \int_0^T B(dX_s, Y_s) + \int_0^T B(X_s, dY_s). \qquad \lesssim \lim_{|\pi|\to 0} \|X\|_\alpha \|Y\|_\beta |\pi|^{\alpha+\beta-1} T = 0$$

the operator norm of B is finite since the finitess dimension

$\square$

$C^{k+\gamma}(\mathbb{R}^d;\mathbb{R})$: is the space of function $f: \mathbb{R}^d \to \mathbb{R}$ s.t. f is k times continuosly diff.

and the k^{th}-order derivative $D^k f$ is locally γ-Hölder continuous, i.e. $\forall K \subset \mathbb{R}^d$ bounded $\exists C$

s.t. $|D^k f(x) - D^k f(y)| \le C|x-y|^\gamma$ $\forall x,y \in K$.

Chain rule for Young Integral: let $X \in C^\alpha([0,T];\mathbb{R}^d)$ and $f \in C^{1+\gamma}_1(\mathbb{R}^d;\mathbb{R})$ for some $\alpha, \gamma \in (0,1]$ s.t. $\alpha(1+\gamma) > 1$. Then $\int_0^T Df(X_u)\, dX_u$ is a well-defined Young integral and:

$$f(X_t) = f(X_0) + \int_0^t Df(X_u)\, dX_u \qquad \forall t \in [0,T].$$

Proof: · Well defined: we want $Df(X) \in C^{\alpha\gamma}$ so that $\alpha + \alpha\gamma = \alpha(1+\gamma) > 1$.

$$|Df(X_t) - Df(X_s)| \underset{\substack{\uparrow \\ Df \in C^{1+\gamma}_1}}{\leq} C\,|X_t - X_s|^\gamma \leq C\,\|X\|_\alpha^\gamma\, |t-s|^{\alpha\gamma}$$

$$\|X\|_\infty \leq |X_0| + T^\alpha \|X\|_\alpha < R \quad \text{so it's bounded and we find } C$$

· Formula: wlog $t = T$:

$$\left| f(X_T) - f(X_0) - \int_0^T Df(X_u)\, dX_u \right| \overset{\text{as in the proof of "formula by parts" + def of Young Integral}}{=} \lim_{|\pi| \to 0} \left| \sum_{[s,t] \in \pi} f(X_t) - f(X_s) - Df(X_s) X_{s,t} \right| = 0$$

$\phi(r) := f(X_s + r X_{s,t})$

$$f(X_t) - f(X_s) = \phi(1) - \phi(0) = \int_0^1 \phi'(r)\, dr = \int_0^1 Df(X_s + r X_{s,t})\, X_{s,t}\, dr$$

$$\Rightarrow |f(X_t) - f(X_s) - Df(X_s) X_{s,t}| = \left| \int_0^1 \big(Df(X_s + r X_{s,t}) - Df(X_s) \big) X_{s,t}\, dr \right| \overset{\text{as previously since } Df \in C^{1+\gamma}_1}{\leq}$$

$$\leq \int_0^1 C\,|r X_{s,t}|^\gamma\, |X_{s,t}|\, dr \underset{\substack{\uparrow \\ 0 \leq r \leq 1}}{\leq} C\,|X_{s,t}|^{1+\gamma} \leq C\,\|X\|_\alpha^{1+\gamma}\, |t-s|^{\alpha(1+\gamma)}$$

$\square$

Associativity of the Young integral: $X \in C^\alpha([0,T], \mathbb{R}^d)$, $K \in C^\beta([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$, $Y \in C^\beta([0,T], \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$ and $\alpha + \beta > 1$. Let $Z = \int_0^\cdot K_u\, dX_u$. Then $Z \in C^\alpha$ and

$$\int_0^T Y_u\, dZ_u = \int_0^T Y_u \circ K_u\, dX_u \qquad \Big(Y \overset{\text{notation for integration}}{\cdot} (K \cdot X) = (Y K) \cdot X \Big)$$

Proof Well-defined: the LHS is well-defined by the fact that the Young int. is a (loc. lip) map from $C^\alpha \times C^\beta \to C^\alpha$; the RHS is well defined because $|Y_t \circ K_t - Y_s \circ K_s| \leq |Y_t - Y_s|\,|K_s| + |Y_s|\,|K_t - K_s| \overset{K,\,Y \text{ bounded}}{\leq} C(\|Y\|_\beta + \|K\|_\beta)\,|t-s|^\beta$

Formula:

$$\int_s^t Y_u\, dZ_u = \underbrace{Y_s Z_{s,t} + O(|t-s|^{\alpha+\beta}) = Y_s K_s X_{s,t} + O(|t-s|^{\alpha+\beta})}_{} = \int_s^t Y_u K_u\, dX_u + O(|t-s|^{\alpha+\beta})$$

we use the estimate of the Young integral: $\int_s^t A_u\, dB_u - Y_s B_{s,t} = O(|t-s|^{\alpha+\beta})$

taking the $\lim_{|\pi| \to 0} \sum_{[s,t] \in \pi}$ on both side we obtain the result. $\quad \sum_{[s,t] \in \pi} O(|t-s|^{\alpha+\beta}) \leq C\,|\pi|^{\alpha+\beta-1}\, T$ $\square$

Controlled paths

We introduce the space of what they will be the valid integrands in the rough integration.

Controlled paths : $\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right]$ and $X \in C^{\alpha}([0,T]; \mathbb{R}^d)$. A pair (Y, Y') is a controlled path w.r.t. X if $Y \in C^{\alpha}([0,T]; \mathbb{R}^m)$, $Y' \in C^{\alpha}([0,T]; \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m))$ and $R^Y \in C_2^{2\alpha}([0,T], \mathbb{R}^m)$

where $R^Y : \Delta_{[0,T]} \longrightarrow \mathbb{R}^m$ is defined implicitly as

← Gubinelli derivative

$$Y_{s,t} = Y_s' X_{s,t} + R^Y_{s,t} \qquad (s,t) \in \Delta_{[0,T]}$$

↖ Remainder

We denote $\mathcal{D}_X^{2\alpha}([0,T]; \mathbb{R}^m)$ the space of controlled paths w.r.t. X.

$\mathcal{D}_X^{2\alpha}$ is a Banach space : $\mathcal{D}_X^{2\alpha}$ equipped with the norm $\|Y, Y'\|_{\mathcal{D}_X^{2\alpha}} = |Y_0| + |Y_0'| + \|Y'\|_{\alpha} + \|R^Y\|_{2\alpha}$ is a Banach space.

we don't need $\|Y\|_{\alpha}$ since $\|Y\|_{\alpha} \leq \|Y'\|_{\infty} \|X\|_{\alpha} + T^{\alpha} \|R^Y\|_{2\alpha}$

Proof : Vector space : if $(Y_1, Y_1'), (Y_2, Y_2') \in \mathcal{D}_X^{2\alpha} \Rightarrow (Y_1 + Y_2, Y_1' + Y_2')$ have the right regularities and the remainder will be $R^{Y_1} + R^{Y_2}$; the same for $(\lambda Y, \lambda Y')$

Complete : Let (Y_n, Y_n') a Cauchy sequence in $\mathcal{D}_X^{2\alpha}$. Then $|(Y_n')_0| + \|Y_n'\|_{\alpha}$ is the C^{α}-norm and by completeness of $C^{\alpha}([0,T]; \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m))$ $\exists\, Y'$ s.t. $Y_n' \xrightarrow{C^{\alpha}} Y'$.

But also $\|\cdot\|_{2\alpha}$ is a real norm the complete space $C_2^{2\alpha}$ so $\exists\, R^Y$ s.t. $R^{Y_n} \overset{C_2^{2\alpha}}{\Longrightarrow} R^Y$.

And obviously $Y_0^n \longrightarrow Y_0$ for some $Y_0 \in \mathbb{R}^d$. But using the def. of remainder:

$$\sup_t |Y_t^n - Y_t^m| \leq |Y_0^n - Y_0^m| + |(Y^n)_0' - (Y^m)_0'| \sup_t |X_{0,t}| + \sup_t |R^{Y^n}_{0,t} - R^{m}_{0,t}|$$

and so Y^n is Cauchy in the uniform norm $\Rightarrow \exists\, Y$ s.t. $Y^n \rightarrow Y$ uniformly

and we know by a proposition in the "C^{α} section" that $Y \in C^{\alpha}$ ($\sup_n Y^n$ is bounded because Y^n is a Cauchy sequence in the sup norm hence bounded). $\square$

Remark : the Gubinelli derivative is not unique : for example if $X, Y \in C^{2\alpha}$, consider Y' continuous then R^Y is automatically 2α-Hölder infact:

$$|R^Y_{s,t}| = |Y_{s,t} - Y_s' X_{s,t}| \leq |Y_{s,t}| + |Y_s' X_{s,t}| \leq \|Y\|_{2\alpha} |t-s|^{2\alpha} + \|Y'\|_{\infty,[0,T]} \|X\|_{2\alpha} |t-s|^{2\alpha}$$

So for every Y' α-Hölder we would have a real Gubinelli derivative.

However if the path is "genuinely rough" in all directions, Y' si uniquenly determined by Y:

<u>Rough at time s</u>: for fixed $s \in [0,T)$ we call $X \in C^{\alpha}([0,T], \mathbb{R}^d)$ "rough at time s" if

$$\forall \; v^* \in (\mathbb{R}^d)^* \setminus \{0\} \quad \text{we have} \quad \lim_{t \downarrow s} \frac{v^*(X_{s,t})}{|t-s|^{2\alpha}} = \infty$$

↑ to see every direction

<u>Truly rough</u>: $X \in C^{\alpha}([0,T], \mathbb{R}^d)$ is truly rough if it is rough on some dense of $[0,T]$.

<u>Uniqueness of Y' under hp. of "Truly rough"</u>: let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^{\alpha}$, $(Y, Y') \in \mathscr{D}_X^{2\alpha}$. Assume X is rough at time $s \in [0,T)$. Then if $Y_{s,t} = O(|t-s|^{2\alpha})$ as $t \downarrow s$ $\Rightarrow Y'_s = 0$. As a consequence, if X is truly rough and $(Y, \tilde{Y}') \in \mathscr{D}_X^{2\alpha}$ is another controlled path $\Rightarrow Y' \equiv \tilde{Y}'$.

↙ we don't need $Y = O(|t-s|^{2\alpha})$!

<u>Proof</u>: <u>1° part</u>: since $(Y, Y') \in \mathscr{D}_X^{2\alpha}$ then

$$Y_{s,t} = Y'_s X_{s,t} + O(|t-s|^{2\alpha}) \quad , \text{ so for } t \in (s, s+\varepsilon)$$

$$\frac{Y'_s X_{s,t}}{|t-s|^{2\alpha}} = \frac{Y_{s,t}}{|t-s|^{2\alpha}} + O(1) \underset{\uparrow}{=} O(1)$$

$Y_{s,t} = O(|t-s|^{2\alpha})$

Now $Y'_s X_{s,t} \in \mathbb{R}^m$, then if $\omega^* \in (\mathbb{R}^m)^*$ we have:

$$\frac{\omega^*(Y'_s X_{s,t})}{|t-s|^{2\alpha}} = \omega^* \left(\frac{Y'_s X_{s,t}}{|t-s|^{2\alpha}} \right) = \omega^*(O(1)) \underset{\uparrow}{=} O(1) \quad \text{as } t \downarrow s$$

ω^* is continuous

Then, we can define $v^* \in (\mathbb{R}^d)^* \setminus \{0\}$ as $v^*(v) := \omega^*(Y'v)$ obtaining

$$\frac{|v^*(X_{s,t})|}{|t-s|^{2\alpha}} = O(1)$$ but X is rough at time s so the RHS

has to be $+\infty$ (and not $O(1)$) unless $v^* = 0$. It means that $\forall \omega^*$ $\forall v$ we have $\omega^*(Y'v) = 0 \implies Y' \equiv 0$.

2° part : we want to show that $Y'_s - \tilde{Y}'_s = 0$ for a rough time s, so that

$Y'_s - Y'_s \equiv 0$ in a dense set of $[0,T]$ and since they are continuous $Y'_s \equiv \tilde{Y}'_s \ \forall s \in [0,T]$.

To see it, we use the first part with $(Z_{s,t} \equiv 0,\ Z'_s = Y'_s - \tilde{Y}'_s)$: infact

$$\begin{cases} Y_{s,t} = Y'_s X_{s,t} + O(|t-s|^{2\alpha}) \\ Y_{s,t} = \tilde{Y}'_s X_{s,t} + O(|t-s|^{2\alpha}) \end{cases} \Longrightarrow \quad 0 = (Y'_s - \tilde{Y}'_s) X_{s,t} + O(|t-s|^{2\alpha}).$$

$\square$

<u>The space of C^K and C^K_b</u> : $f \in C^K$ whenever f is K times continuously differentiable, $f \in C^K_b$ when additionally f and all its derivatives up to order K are uniformly bounded. We set:

$$\|f\|_{C^K_b} := \|f\|_\infty + \|Df\|_\infty + \dots + \|D^K f\|_\infty$$

<u>$(f(X), Df(X))$ as controlled path</u> : $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and $X \in C^\alpha$. If $f \in C^2$ then

$(f(X), Df(X)) \in \mathcal{D}^{2\alpha}_X$.

<u>Proof</u> : • $f(X) \in C^\alpha([0,T]; \mathbb{R}^d)$:

$$|f(X_t) - f(X_s)| = \left| \int_0^1 Df(X_s + \theta(X_t - X_s))(X_t - X_s)\, d\theta \right|$$

$$\le M_1 |X_t - X_s| = O(|t-s|^\alpha)$$

X is continuous $\Rightarrow$
$X([0,T])$ is compact
$\Rightarrow Df(\mathrm{Im}\, X) = M_1 < +\infty$

• $Df(X) \in C^\alpha([0,T]; \mathbb{R}^d)$:

Attention: $Df(X_t) \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m) \Rightarrow \int_0^1 D^2 f(X_s + \theta(X_t - X_s))[X_t - X_s]\, d\theta \in \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m)$ and infact the only reasonable def. is $D^2 f(v)[w] : \mathbb{R}^d \to \mathbb{R}^m$
$D^2 f(v)[w](u) = u^T D^2 f(v) w =: D^2 f(v)[w, u]$

$$|Df(X_t) - Df(X_s)| = \left| \int_0^1 D^2 f(X_s + \theta(X_t - X_s))[X_t - X_s]\, d\theta \right|$$

there can't be 2α

$$\le M_2 |X_t - X_s| = O(|t-s|^\alpha)$$

• The remainder : we set $R^{f(X)}_{s,t} := f(X_t) - f(X_s) - Df(X_s) X_{s,t}$ and so:

$$|f(X_t) - f(X_s) - Df(X_s) X_{s,t}| = \left| \int_0^1 (Df(X_s + \theta X_{s,t}) - Df(X_s)) X_{s,t}\, d\theta \right|$$

$$= \left| \int_0^1 \int_0^u D^2 f(X_s + u X_{s,t})[X_{s,t}, X_{s,t}]\, du\, d\theta \right| \leq$$

$$\leq M_2 |X_{s,t}|^2 \leq M_2 \|X\|_\alpha^2 |t-s|^{2\alpha}$$

here $(Df(X_s + \theta X_{s,t}) - Df(X_s))X_s$ is a vector and so the integral becomes $\int_0^u X_{s,t}^T D^2 f(X_s + u X_{s,t}) X_{s,t} du$

and so we have the factor $|X_{s,t}|^2$

$\square$

The product of controlled path is a controlled path : let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, $X \in C^\alpha([0,T], \mathbb{R}^d)$, $Y \in D_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$, $Z \in D_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^n))$. Then $(YZ, (YZ)') \in D_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^n))$

$$(YZ)' = YZ' + Y'Z.$$

← Using the indexes: (see below for the "contraction rules")

$$((YZ)')^i_{jk} = Y^i_\ell Z^\ell_{jk} + Y'^i_{j\ell} Z^\ell_k$$

Proof : The regularity of YZ is ok, we just need to control the regularity of the remainder:

$$R^{YZ}_{s,t} := (YZ)_{s,t} - (YZ)'_s X_{s,t} = Y_t Z_t - Y_s Z_s - Y_s Z'_s X_{s,t} - Y'_s Z_s X_{s,t} =$$

$$= \left(Y_s R^Z_{s,t} - Y_s R^Z_{s,t}\right) + \left(R^Y_{s,t} Z_s - R^Y_{s,t} Z_s\right) + Y_t Z_t - Y_s Z_s - Y_s Z'_s X_{s,t} - Y'_s Z_s X_{s,t} =$$

$$= Y_s R^Z_{s,t} + R^Z_{s,t} Z_s + Y_{s,t} Z_{s,t}$$

$$\Rightarrow |R^{Y,Z}_{s,t}| \leq \|Y\|_\infty \|R_{s,t}\|_{2\alpha} |t-s|^{2\alpha} + \|Z\|_\infty \|R^Z_{s,t}\|_{2\alpha} |t-s|^{2\alpha} + \|Y\|_\alpha \|Z\|_\alpha |t-s|^{2\alpha}$$

$\square$

Rough Integration

Rough integral : let $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and let $\mathbf{X} := (X, \mathbb{X}) \in C^\alpha([0,T], \mathbb{R}^d)$ be a rough path.

Let $(Y, Y') \in D_X^{2\alpha}([0,T]; \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m))$ be a controlled path and let R^Y be the corresponding remainder term. Then the limit

they're well defined! $Y'_u \in \mathcal{L}(\mathbb{R}^d; \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m)) \simeq \mathcal{L}(\mathbb{R}^d \otimes \mathbb{R}^d, \mathbb{R}^m)$

$$\int_0^t Y_u \, d\mathbf{X}_u := \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_u X_{u,v} + Y'_u \mathbb{X}_{u,v} \qquad (\pi \text{ partition of } [0,t])$$

This limit is called the rough integral of (Y, Y') against $\mathbf{X}$ which comes with the estimate

$$\left| \int_0^t Y_u \, d\mathbf{X}_u - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \leq C \left(\|R^Y\|_{2\alpha} \|X\|_\alpha + \|Y'\|_\alpha \|\mathbb{X}\|_{2\alpha} \right) |t-s|^{3\alpha}$$

$\forall (s,t) \in \Delta_{[0,T]}$ and C that depends only on α.

<u>Proof</u>: set $A_{s,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t}$, if we prove that

$$|\delta A_{s,u,t}| \leq \left(\|R^Y\|_{2\alpha} \|X\|_\alpha + \|Y'\|_\alpha \|\mathbb{X}\|_{2\alpha} \right) |t-s|^{3\alpha}$$ we have finished

for the sewing lemma ($3\alpha > 1$).

So :

$$\delta A_{s,u,t} = A_{s,t} - A_{s,u} - A_{u,t} = Y_s X_{s,t} + Y'_s \mathbb{X}_{s,t} - Y_s X_{s,u} - Y'_s \mathbb{X}_{s,u} - Y_u X_{u,t} - Y'_u \mathbb{X}_{u,t}$$

Y'_s linear $\rightarrow$ $$= \underbrace{Y_s X_t - Y_s X_u - Y_u X_{u,t}}_{-Y_{s,u} X_{u,t}} + Y'_s \left(\mathbb{X}_{s,t} - \mathbb{X}_{s,u} \right) - Y'_u \mathbb{X}_{u,t}$$

$\mathbb{X}_{s,t} = \mathbb{X}_{s,u} + \mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}$ $\rightarrow$ $$= -Y_{s,u} X_{u,t} + Y'_s \left(\mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t} \right) - Y'_u \mathbb{X}_{u,t}$$

Y'_s linear $\rightarrow$ $$= -Y_{s,u} X_{u,t} + Y'_s \mathbb{X}_{u,t} + Y'_s (X_{s,u} \otimes X_{u,t}) - Y'_u \mathbb{X}_{u,t}$$

$\mathcal{L}(\mathbb{R}^d ; \mathcal{L}(\mathbb{R}^d ; \mathbb{R}^m)) \simeq \mathcal{L}(\mathbb{R}^d \otimes \mathbb{R}^d ; \mathbb{R}^m)$ using the identification $\tilde{\varphi}(v \otimes w) = \varphi(v)(w)$

$$= -Y_{s,u} X_{u,t} + (Y'_s(X_{s,u}))(X_{u,t}) - Y'_{s,u} \mathbb{X}_{u,t}$$

$$= (Y'_s X_{s,u} - Y_{s,u})(X_{u,t}) - Y'_{s,u} \mathbb{X}_{u,t} = -R^Y_{s,u}(X_{u,t}) - Y'_{s,u} \mathbb{X}_{u,t}$$

Hence $|\delta A_{s,u,t}| \leq \left(\|R^Y\|_{2\alpha} \|X\|_\alpha + \|Y'\|_\alpha \|\mathbb{X}\|_{2\alpha} \right) |t-s|^{3\alpha}$.

<u>Integral formula for rough paths with different lifts</u> : $\alpha \in \left(\frac{1}{3}, \frac{1}{2} \right]$, let $F \in C^{2\alpha}([0,T], \mathbb{R}^d \times \mathbb{R}^d)$ ($\simeq \mathbb{R}^d \otimes \mathbb{R}^d$) a 2α-Hölder path and let $\mathbf{X} = (X, \mathbb{X})$, $\tilde{\mathbf{X}} = (X, \tilde{\mathbb{X}}) \in \mathcal{C}^\alpha$ s.t. $\tilde{\mathbb{X}}_{s,t} = \mathbb{X}_{s,t} + F_{s,t}$

Let $(Y, Y') \in D_X^{2\alpha}$, then :

$$\int_0^T Y_u \, d\tilde{\mathbf{X}}_u = \int_0^T Y_u \, d\mathbf{X}_u + \int_0^T Y'_u \, dF_u$$ Young Integral!

<u>Proof</u> : · <u>Well-defined</u> : $Y' \in C^\alpha([0,T] ; \mathcal{L}(\mathbb{R}^d \otimes \mathbb{R}^d ; \mathbb{R}^m))$ $F \in C^{2\alpha}([0,T] ; \mathbb{R}^d \times \mathbb{R}^d)$ so

$2\alpha + \alpha = 3\alpha > 1$ and the Young integral is well defined

· Formula: $\int_0^T Y_u \, d\tilde{\mathbf{X}}_u = \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} Y_s \tilde{X}_{s,t} + Y_s' \tilde{\mathbb{X}}_{s,t} =$

$$= \lim_{|\pi|\to 0} \sum_{[s,t]\in\pi} Y_s X_{s,t} + Y_s' \mathbb{X}_{s,t} + Y_s' F_{s,t}$$

$$= \int_0^T Y_u \, d\mathbf{X}_u + \int_0^T Y_u' \, dF_u .$$

$\square$

$\left(\int_0^\cdot Y_u \, d\mathbf{X}_u, Y\right) \in D_X^{2\alpha}$: if $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha$ and $(Y, Y') \in D_X^{2\alpha}([0,T]; \mathcal{L}(\mathbb{R}^d; \mathbb{R}^m))$

$$\left(\int_0^\cdot Y_u \, d\mathbf{X}_u, Y\right) \in D_X^{2\alpha}([0,T]; \mathbb{R}^m)$$

Proof : · $\int_0^\cdot Y_u \, d\mathbf{X}_u \in C^\alpha([0,T]; \mathbb{R}^m)$: from the estimate in the theorem we have

$$\left| \int_0^\cdot Y_u \, d\mathbf{X}_u \right| \le |Y_s X_{s,t}| + |Y_s' \mathbb{X}_{s,t}| + O(|t-s|^{3\alpha})$$

$$\le \|Y\|_\infty \|X\|_\alpha |t-s|^\alpha + \|Y'\|_\infty \|\mathbb{X}_{s,t}\|_{2\alpha} |t-s|^{2\alpha} + O(|t-s|^{3\alpha})$$

· $Y \in C^\alpha([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$: by def. of $\int_0^\cdot Y_u \, d\mathbf{X}_u$ we have $(Y, Y') \in D_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$.

· The remainder : same estimate used in the first point

$$\left| R_X^{\int_0^\cdot Y_u d\mathbf{X}_u} \right| := \left| \int_s^t Y_u \, d\mathbf{X}_u - Y_s X_{s,t} \right| \le \|Y'\|_\infty \|\mathbb{X}\|_{2\alpha} |t-s|^{2\alpha} + O(|t-s|^{3\alpha})$$

$\square$

Stability of the remainders of the integrals : let $\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right]$, and let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha$ and $\tilde{\mathbf{X}} = (\tilde{X}, \tilde{\mathbb{X}}) \in \mathscr{C}^\alpha$ be rough paths. Let $(Y, Y') \in D_X^{2\alpha}$ and $(\tilde{Y}, \tilde{Y}') \in D_{\tilde{X}}^{2\alpha}$ controlled paths, and R^Y, $R^{\tilde{Y}}$ the corresponding remainder terms. Then, $\exists C$ constant depending on α and T s.t. :

they are operator norm

· $\|Y - \tilde{Y}\|_\alpha \le C\left(\left(|Y_0' - \tilde{Y}_0'| + \|Y' - \tilde{Y}'\|_\alpha\right) \|X\|_\alpha + \left(|\tilde{Y}_0'| + \|\tilde{Y}'\|_\alpha\right) \|X - \tilde{X}\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha} T^\alpha \right)$

· $\left\| R^{\int_0^\cdot Y_u d\mathbf{X}_u} - R^{\int_0^\cdot \tilde{Y}_u d\tilde{\mathbf{X}}_u} \right\|_{2\alpha} \le C\left(\left(|Y_0' - \tilde{Y}_0'| + \|Y' - \tilde{Y}'\|_\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha}\right) \|\mathbf{X}\|_\alpha + \left(|\tilde{Y}_0'| + \|\tilde{Y}'\|_\alpha + \|R^{\tilde{Y}}\|_{2\alpha}\right) \|\mathbf{X}; \tilde{\mathbf{X}}\|_\alpha \right)$

Proof :

· 1st inequality :

$$|Y_{s,t} - \tilde{Y}_{s,t}| = |Y_s' X_{s,t} + R^Y_{s,t} - \tilde{Y}_s' \tilde{X}_{s,t} - R^{\tilde{Y}}_{s,t}|$$

$$+\tilde{Y}_s' X_{s,t} - \tilde{Y}_s' X_{s,t} \leq |Y_s' - \tilde{Y}_s'|\,|X_{s,t}| + |\tilde{Y}_s'|\,|X_{s,t} - \tilde{X}_{s,t}| + |R^Y_{s,t} - R^{\tilde{Y}}_{s,t}|$$

$$\leq \left(\|Y' - \tilde{Y}'\|_\infty \|X\|_\alpha + \|\tilde{Y}\|_\infty \|X - \tilde{X}\|_\alpha\right)|t-s|^\alpha + \|R^Y - R^{\tilde{Y}}\|_{2\alpha}|t-s|^{2\alpha}$$

Now using $|t-s|^\alpha \leq T^\alpha$ on the last term and $\|V\|_\infty \leq |V_0| + T^\alpha \|V\|_\alpha$ we have the result using a new constant C that incorporates the T^α factors.

<u>2° inequality</u>: from the previous prop we have

$$\left|R^{\int_0^\cdot Y_u d\mathbf{X}_u} - R^{\int_0^\cdot \tilde{Y}_u d\tilde{\mathbf{X}}_u}\right| = \underbrace{\left|\int_s^t Y_u \, d\mathbf{X}_u - Y_s X_{s,t} - \int_s^t \tilde{Y}_u \, d\tilde{\mathbf{X}}_u + Y_s \tilde{X}_{s,t}\right|}_{\text{as in the case of the Young integral we want to estimate this with the sewing lemma}} \quad (\star)$$

So $A_{s,t} := Y_s X_{s,t} + Y' \mathbb{X}_{s,t}$, $\tilde{A}_{s,t} := \tilde{Y}_s \tilde{X}_{s,t} + \tilde{Y}' \tilde{\mathbb{X}}_{s,t}$, $\Delta_{s,t} := A_{s,t} - \tilde{A}_{s,t}$:

$$\delta \Delta_{s,u,t} = \delta A_{s,t} - \delta \tilde{A}_{s,t} \underset{\substack{\uparrow \\ \text{see proof of Rough int. theorem}}}{=} -\left(R^Y_{s,u}(X_{u,t}) + Y'_{s,u}\mathbb{X}_{u,t} - R^{\tilde{Y}}_{s,u}\tilde{X}_{u,t} - \tilde{Y}'_{s,u}\tilde{\mathbb{X}}_{u,t}\right)$$

Hence:

$$|\delta \Delta_{s,u,t}| = |R^Y_{s,u} X_{u,t} + Y'_{s,u}\mathbb{X}_{u,t} - R^{\tilde{Y}}_{s,u}\tilde{X}_{u,t} - \tilde{Y}'_{s,u}\tilde{\mathbb{X}}_{u,t}|$$

sum and subtract $R^{\tilde{Y}}_{s,u} X_{u,t}$ and $\tilde{Y}'_{s,u}\mathbb{X}_{u,t}$

$$\to \leq |R^Y_{s,u} - R^{\tilde{Y}}_{s,u}|\,|X_{u,t}| + |R^{\tilde{Y}}_{s,u}|\,|X_{u,t} - \tilde{X}_{u,t}| + |Y'_{s,u} - \tilde{Y}'_{s,u}|\,|\mathbb{X}_{u,t}| + |\tilde{Y}'_{s,u}|\,|\mathbb{X}_{u,t} - \tilde{\mathbb{X}}_{u,t}|$$

op. norm

$$\leq \underbrace{\left(\|R^Y - R^{\tilde{Y}}\|_{2\alpha}\|X\|_\alpha + \|R^{\tilde{Y}}\|_{2\alpha}\|X - \tilde{X}\|_\alpha + \|Y' - \tilde{Y}'\|_\alpha \|\mathbb{X}\|_{2\alpha} + \|\tilde{Y}'\|_\alpha \|\mathbb{X} - \tilde{\mathbb{X}}\|_{2\alpha}\right)|t-s|^{3\alpha}}_{(*_1)}.$$

So by the sewing lemma $\exists\, \gamma$ and C s.t.

$$|\gamma_t - \gamma_s - \Delta_{s,t}| \leq C \cdot (*_1) \qquad (*_2)$$

and

$$\gamma_t - \gamma_s = \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} \Delta_{u,v} = \lim_{|\pi| \to 0}\left(\sum_{[u,v]\in\pi} A_{u,v} - \sum_{[u,v]\in\pi} \tilde{A}_{u,v}\right) = \int_s^t Y_u d\mathbf{X}_u - \int_s^t \tilde{Y}_u \tilde{\mathbf{X}}_u$$

So that rewriting $(*_2)$:

$$\left|\int_s^t Y_u d\mathbf{X}_u - \int_s^t \tilde{Y}_u \tilde{\mathbf{X}}_u - \left(Y_s X_{s,t} + Y_s' \mathbb{X}_{s,t} - \tilde{Y}_s \tilde{X}_{s,t} - \tilde{Y}_s' \tilde{\mathbb{X}}_{s,t}\right)\right| \leq C \cdot (*_1) \qquad (*_3)$$

but

$$|Y_s' \mathbb{X}_{s,t} - \tilde{Y}_s' \tilde{\mathbb{X}}_{s,t}| \leq |Y_s' - \tilde{Y}_s'|\,|\mathbb{X}_{s,t}| + |\tilde{Y}_s'|\,|\mathbb{X}_{s,t} - \tilde{\mathbb{X}}_{s,t}| \leq$$

$$\leq \left(\|Y' - \tilde{Y}'\|_\infty \|\mathbb{X}\|_{2\alpha} + \|Y'\|_\infty \|\mathbb{X} - \mathbb{X}\|_{2\alpha}\right)(t-s)^{2\alpha} \qquad (*_4)$$

So by using $(\star_4)$ in $(\star_3)$ we obtain

$$(\star) \le \left(\| Y' - \tilde{Y}' \|_\infty \, \| \mathbb{X} \|_{2\alpha} + \| Y' \|_\infty \, \| \mathbb{X} - \tilde{\mathbb{X}} \|_{2\alpha} \right) (t-s)^{2\alpha} + C \cdot (\star_1)$$

$$\text{So } \left\| R^{\int_0^\cdot Y_u \, d u} - R^{\int_0^\cdot \tilde{Y}_u \, d u} \right\|_{2\alpha} \le \| Y' - \tilde{Y}' \|_\infty \, \| \mathbb{X} \|_{2\alpha} + \| Y' \|_\infty \, \| \mathbb{X} - \tilde{\mathbb{X}} \|_{2\alpha} + C (\star_1)$$

Then we finish exactly as in the first inequality

<u>Corollary (stability of the rough integral)</u> : Under the same assumptions of above $\exists\, C$ constant depending on α and T s.t.

$$\left\| \int_0^\cdot Y_u \, d\mathbf{X}_u - \int_0^\cdot \tilde{Y}_u \tilde{\mathbf{X}}_u \right\|_\alpha \le C \left(1 + \| X \|_\alpha + \| \tilde{X} \|_\alpha \right) \left(\| \tilde{Y}, \tilde{Y}' \|_{D_X^{2\alpha}} \| \mathbf{X} ; \tilde{\mathbf{X}} \|_\alpha + \left(|Y_0 - \tilde{Y}_0| + |Y_0' - \tilde{Y}_0'| + \| Y' - \tilde{Y}' \|_\alpha + \| R^Y - R^{\tilde{Y}} \|_{2\alpha} \right) \| \mathbf{X} \|_\alpha \right)$$

<u>Proof</u> : we use the previous theorem with the controlled paths $\left(\int_0^\cdot Y_u \, d\mathbf{X}_u , Y \right)$ and $\left(\int_0^\cdot \tilde{Y}_u \, d\tilde{\mathbf{X}}_u , \tilde{Y} \right)$. By the 1^{st} inequality of the theorem we get an inequality for $\int \cdot - \int \tilde{\cdot}$ where they appear $\| Y - \tilde{Y} \|$ (that we estimate another time with the 1^{st} inequality) and $\| R^{\int \cdot} - R^{\int \tilde{\cdot}} \|$ that we estimate with the second inequality.

Now, we have only the terms that we want and we just need to use the right constant. $\square$

Integrate controlled paths against each other

<u>Idea</u> : two controlled paths of X are both "similar" to X, so they look similar to each other and maybe the integration against each other makes sense.

<u>Integral of a controlled path against another one</u> : let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0,T] ; \mathbb{R}^d)$ be a rough path, let $(Y, Y') \in D_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m))$ and $(Z, Z') \in D_X^{2\alpha}([0,T], \mathbb{R}^n)$. Then the limit

$$\int_0^t Y_u \, dZ_u := \lim_{|\pi| \to 0} \sum_{[s,t] \in \pi} Y_s Z_{s,t} + Y_s' Z_s' \mathbb{X}_{s,t} \qquad (\pi \text{ partition of } [0,t])$$

exists $\forall t \in [0,T]$. Moreover, we have the estimate

$$\left| \int_s^t Y_u \, dZ_u - Y_s Z_{s,t} - Y_s' Z_s' \mathbb{X}_{s,t} \right| \le C \left(\| Y' \|_\infty \| Z' \|_\alpha \| X \|_\alpha^2 + \| Y \|_\alpha \| R^Z \|_{2\alpha} + \| R^Y \|_{2\alpha} \| Z' \|_\infty \| X \|_\alpha + \| Y' Z' \|_\alpha \| \mathbb{X} \|_{2\alpha} \right) |t-s|^{3\alpha}$$

$\rightarrow$ Remark on the limit : to understand $Y_s Z_{s,t} + Y_s' Z_s' \mathbb{X}_{s,t}$ we have to work with the tensors : when we find a domain the index is "down", when we find a codomain the index is up :

- $Y_s \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) \rightsquigarrow (Y_s)^i_j$ $\leftarrow$ we will use i for indexes of "dimension" m, j for n, k for d
- $Z_{s,t} \in \mathbb{R}^n \rightsquigarrow (Z_{s,t})^j$ (the vector is always "up")
- $Y_s' \in \mathcal{L}(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)) \rightsquigarrow (Y_s')^i_{k,j}$
- $Z_s' \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^n) \rightsquigarrow (Z_s')^j_k$
- $\mathbb{X}_{s,t} \in \mathbb{R}^d \times \mathbb{R}^d \rightsquigarrow (\mathbb{X}_{s,t})^{k,k}$ so

$$(Y_s Z_{s,t} + Y_s' Z_s' \mathbb{X}_{s,t})^i = (Y_s)^i_j (Z_{s,t})^j + (Y_s')^i_{k,j} (Z_s')^j_k (\mathbb{X}_{s,t})^{k,k} \leftarrow$$

$$= (Y_s)^i_j (Z_{s,t})^j + (Y_s')^i_{k,j} (Z_s')^j_\ell (\mathbb{X}_{s,t})^{k,\ell}$$

(we have to contract the indexes with same dimension, use another index ℓ for dim. d)

$\rightarrow$ Proof : We want to use the sewing lemma: $A_{s,t} := Y_s Z_{s,t} + Y_s' Z_s' \mathbb{X}_{s,t}$.

$$\delta A_{s,u,t} = Y_s Z_{s,t} + Y_s' Z_s' \mathbb{X}_{s,t} - Y_s Z_{s,u} - Y_s' Z_s' \mathbb{X}_{s,u} - Y_u Z_{u,t} - Y_u' Z_u' \mathbb{X}_{u,t}$$

$$= Y_s Z_{u,t} + Y_s' Z_s' (\mathbb{X}_{s,t} - \mathbb{X}_{s,u}) - Y_u Z_{u,t} - Y_u' Z_u' \mathbb{X}_{u,t}$$

Chen's relation

$$= Y_s Z_{u,t} + Y_s' Z_s' (\mathbb{X}_{u,t} + X_{s,u} \otimes X_{u,t}) - Y_u Z_{u,t} - Y_u' Z_u' \mathbb{X}_{u,t}$$

$$= -Y_{s,u} Z_{u,t} - (Y'Z')_{s,u} \mathbb{X}_{u,t} + Y_s' Z_s' X_{s,u} \otimes X_{u,t}$$

Y, Z controlled by X

$$= -(\underbrace{Y_s' X_{s,u} + R^Y_{s,u}}_{Y_{s,u}})(Z_u' X_{u,t} + R^Z_{u,t}) - (Y'Z')_{s,u} \mathbb{X}_{u,t} + Y_s' Z_s' X_{s,u} \otimes X_{u,t}$$

$$= -(Y_s' X_{s,u})(Z_u' X_{u,t}) - R^Y_{s,u} Z_u' X_{u,t} - Y_{s,u} R^Z_{u,t} - (Y'Z')_{s,u} \mathbb{X}_{u,t}$$

$$+ Y_s' Z_s' X_{s,u} \otimes X_{u,t}$$

$(Y_s' X_{s,u})(Z_u' X_{u,t})$
$= Y_s' Z_u' X_{s,u} \otimes X_{u,t}$

$$= - Y_s' Z'_{s,u} X_{s,u} \otimes X_{u,t} - R^Y_{s,u} Z_u' X_{u,t} - Y_{s,u} R^Z_{u,t} - (Y'Z')_{s,u} \mathbb{X}_{u,t}$$

$$\Rightarrow |\delta A_{s,u,t}| \leq \left(\|Y\|_\infty \|Z'\|_\alpha \|X\|_\alpha^2 - \|R^Y\|_{2\alpha} \|Z\|_\infty \|X\|_{2\alpha} - \|Y\|_\alpha \|R^Z\|_{2\alpha} - \|YZ\|_\alpha \|\mathbb{X}\|_{2\alpha}\right)|t-s|^{3\alpha}$$

let us this equality with indexes : $((Y_s' X_{s,u})(Z_u' X_{u,t}))^i = (Y_s')^i_{k,j} X^k_{s,u} (Z_u')^j_\ell X^\ell_{u,t} =$

$$= (Y_s')^i_{k,j} (Z_u')^j_\ell X^k_{s,u} X^\ell_{u,t} =$$

$$= (Y_s' Z_u' X_{s,u} \otimes X_{u,t})^i$$

$\square$

Associativity of the rough integral: let $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T], \mathbb{R}^d)$ be a rough path. Now:

- $(K, K') \in \mathcal{D}_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^d))$ $\left(\Rightarrow \int K \, d\mathbf{X} \text{ is well-defined and } (Z, Z') := \right.$ $\left. := \left(\int K d\mathbf{X}, K\right) \in \mathcal{D}_X^{2\alpha}([0,T], \mathbb{R}^d)\right)$
- $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m))$ $\Rightarrow$ (so that $\int K \, dZ$ is well defined for the previous prop.)

Then:

$$\int_0^\cdot Y_u \, dZ_u = \int_0^\cdot Y_u K_u \, d\mathbf{X}_u$$

by the prop. on the product, $Y_u K_u \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)$ and so the rough integral makes sense

Proof:

$$\int_s^t Y_u \, dZ_u \overset{\text{previous theorem}}{=} Y_s Z_{s,t} + Y_s' Z_s' \mathbb{X}_{s,t} + O(|t-s|^{3\alpha})$$

same property for $Z_{s,t}$ and $Z' = K$ →

$$= Y_s \left(K_s X_{s,t} + K_s' \mathbb{X}_{s,t} + O(|t-s|^{3\alpha})\right) + Y_s' K_s \mathbb{X}_{s,t} + O(|t-s|^{3\alpha})$$

$$= Y_s K_s X_{s,t} + (Y_s K_s' + Y_s' K_s) \mathbb{X}_{s,t} + O(|t-s|^{3\alpha}) =$$

property of the product →

$$= Y_s K_s X_{s,t} + (YK)_s' \mathbb{X}_{s,t} + O(|t-s|^{3\alpha})$$

$$= \int_s^t Y_u K_u \, d\mathbf{X}_u + O(|t-s|^{3\alpha})$$

So taking the $\lim_{|\pi| \to 0} \sum_{[s,t] \in \pi}$ on both sides we obtain $\int_0^T Y_u \, dZ_u = \int_0^T Y_u K_u \, d\mathbf{X}_u$

$\left(\sum_{[s,t] \in \pi} O(|t-s|^{3\alpha}) \leq C \, |\pi|^{3\alpha - 1} T \right)$. $\square$

Integral of a controlled path against itself: $\mathbf{X} = (X, \mathbb{X}) \in \mathscr{C}^\alpha([0,T]; \mathbb{R}^d)$, $(Z, Z') \in \mathcal{D}_X^{2\alpha}([0,T]; \mathbb{R}^n)$

We can well define

$$\mathbb{R}^n \otimes \mathbb{R}^n \ni \int_s^t Z_{s,r} \otimes dZ_r := \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Z_u \otimes Z_{u,v} + (Z_u' \otimes Z_u') \mathbb{X}_{u,v}$$

we are identifying $\mathcal{L}(\mathbb{R}^d, \mathbb{R}^n) \otimes \mathcal{L}(\mathbb{R}^d, \mathbb{R}^n) \cong \mathcal{L}(\mathbb{R}^d \otimes \mathbb{R}^d, \mathbb{R}^n \otimes \mathbb{R}^n)$ using $(f \otimes g)(v \otimes w) = f(v) \otimes g(w)$

Proof: consider the prop. "Integral of controlled path against another one". We have the same (Z, Z'), then we define $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^n, \underbrace{\mathbb{R}^n \otimes \mathbb{R}^n}_{:= \mathbb{R}^m}))$ as

$Y \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n \otimes \mathbb{R}^n)$, $Y_t(v) := Z_t \otimes v$ and

$Y'_t \in \mathcal{L}(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n \otimes \mathbb{R}^n))$ $\quad Y'_t(\underset{\mathbb{R}^d}{a})(\underset{\mathbb{R}^n}{b}) := \underset{\mathcal{L}(\mathbb{R}^d, \mathbb{R}^n)}{Z'_t}(a) \otimes b \in \mathbb{R}^n \otimes \mathbb{R}^n$ $\quad (Y'_t)_i$

The regularities hold (it's a computation), so using the prop. we know it exists:

$$\lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Y_u(Z_{u,v}) + Y'_u Z'_u \mathbb{X}_{u,v} = \left(Y'_u Z'_u\right)(\underset{\mathbb{R}^d \otimes \mathbb{R}^d}{v \otimes w}) \overset{\text{the only way to make sense}}{=}$$

$$= Y'_u(v)\left(Z'_u(w)\right) \underset{\text{def of } Y'_u}{=} Z'_u(v) \otimes Z'_u(w)$$

$$= \lim_{|\pi| \to 0} \sum_{[u,v] \in \pi} Z_u \otimes Z_{u,v} + \left(Z'_u \otimes Z'_u\right)\left(\mathbb{X}_{u,v}\right)$$

$\square$

Controlled path as rough path : given a rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0,T], \mathbb{R}^d)$ and $(Z, Z') \in \mathcal{D}_X^{2\alpha}([0,T], \mathbb{R}^m)$, we can define:

$$\mathbb{Z}_{s,t} := \int_s^t Z_{s,r} \otimes dZ_r \quad \leftarrow \text{using the previous definition}$$

Then, $(Z, \mathbb{Z}_{s,t}) \in \mathcal{C}^\alpha([0,T], \mathbb{R}^m)$.

Proof : Chen's relation:

$$\mathbb{Z}^{i,j}_{s,t} = \int_s^t Z^i_{s,r}\, dZ^j_r \underset{\substack{\text{the sewing lemma gives us always}\\ \text{an additive integral}}}{=} \int_s^u Z^i_{s,r}\, dZ^j_r + \int_u^t Z^i_{s,r}\, dZ^j_r = \mathbb{Z}_{s,u} + Z^i_{s,u} Z^j_{u,t} + \mathbb{Z}_{u,t}.$$

$\pm Z^i_u$

Regularity : Use the boundary given by the prop. "Integral of controlled path against another one" where we have

$$\left| \int_s^t Z_{s,r} \otimes dZ_r - \underbrace{Z_s \otimes Z_{s,t} - (Z'_s \otimes Z'_s)\, \mathbb{X}_{s,t}}_{\text{this terms have } 2\alpha\text{-höl. regularity}} \right| \leq O(|t-s|^{3\alpha}) \quad \text{so}$$

$$\left| \int_s^t Z_{s,r} \otimes dZ_r \right| \leq O(|t-s|^{2\alpha}) + O(|t-s|^{3\alpha}) \leq O(|t-s|^{2\alpha})$$

$\square$

Consistency between the integral of a controlled path against a controlled path seen as a controlled path and seen as a rough path :

- $\mathbf{X} := (X, \mathbb{X}) \in \mathcal{C}^\alpha([0,T], \mathbb{R}^d)$ rough path.
- $(Z, Z') \in \mathcal{D}_X^{2\alpha}([0,T], \mathbb{R}^m)$ and we define $\mathbf{Z} := (Z, \mathbb{Z}) \in \mathcal{C}^\alpha([0,T], \mathbb{R}^m)$ as seen before
- $(Y, Y') \in \mathcal{D}_Z^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$

Then $(Y, Y'Z') \in \mathcal{D}_X^{2\alpha}([0,T], \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$ and we have:

$$\int_0^\cdot Y_u \, d\mathbf{Z}_u = \int_0^\cdot Y_u \, dZ_u$$

the integral of the Z-controlled path (Y, Y') against the rough path $(Z, \mathbb{Z})$

The integral of the X-controlled path $(Y, Y'Z')$ against another X-controlled path (Z, Z')

Remark: $Y'Z'$ must be in $\mathcal{L}(\mathbb{R}^d, \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$, now $Z' \in \mathcal{L}(\mathbb{R}^d, \mathbb{R}^m)$ while $Y' \in \mathcal{L}(\mathbb{R}^m, \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$ (is controlled by Z!) so

$$(Y'Z')(\underset{\mathbb{R}^d}{v})(\underset{\mathbb{R}^m}{w}) = \underbrace{Y'(\underbrace{Z'(v)}_{\mathbb{R}^m})}_{\mathcal{L}(\mathbb{R}^m, \mathbb{R}^n)}(w) \;] \in \mathbb{R}^n$$

Proof: Regularity of the remainder ($(Y, Y'Z') \in \mathcal{D}_X^{2\alpha}$):

the remainder is $Y_{s,t} - Y_s' Z_s' X_{s,t} = Y_{s,t} - Y_s'(Z_{s,t} - R^X_{s,t}) = R^Y_{s,t} + Y_s' R^X_{s,t}$

$$Z_{s,t} = Z_s' X_{s,t} + R^X_{s,t} \qquad Y_{s,t} = Y_s' Z_{s,t} + R^Y_{s,t}$$

But the RHS 2α-Hölder continuous: $|R^Y_{s,t} + Y_s' R^X_{s,t}| \le \|R^Y_{s,t}\|_{2\alpha} |t-s|^{2\alpha} + \|Y_s'\|_\infty \|R^X_{s,t}\|_{2\alpha} |t-s|^{2\alpha}$.

Equality: Using all the estimates seen:

estimate on the def of rough integral

$$\int_s^t Y_u \, d\mathbf{Z}_u = Y_s Z_{s,t} + Y' \mathbb{Z}_{s,t} + O(|t-s|^{3\alpha})$$

$\mathbb{Z}_{s,t} = Z_{s,t} \otimes Z_{s,t} + (Z_s' \otimes Z_s')\mathbb{X}_{s,t} + o(|t-s|^{3\alpha})$

$$= Y_s Z_{s,t} + Y'(Z_s' \otimes Z_s')\mathbb{X}_{s,t} + O(|t-s|^{3\alpha})$$

so using $Z_{s,t} = Z_s' X_{s,t} + R^Z_{s,t}$ and Chen we see that the 1st part is $O(|t-s|^{3\alpha})$

$Y_s' Z_s' Z_s' (a \otimes b) = Y_s'(Z_s'(a))(Z_s'(b))$

$$= Y_s Z_{s,t} + Y_s' Z_s' Z_s' \mathbb{X}_{s,t} + O(|t-s|^{3\alpha})$$

$$= \int_s^t Y_u \, dZ_u + O(|t-s|^{3\alpha})$$

def. of integration controlled path against another one

So taking the $\lim_{|\pi| \to 0} \sum_{[s,t] \in \pi}$ on both sides we obtain the thesis as always. $\square$